# Does the Crowd Add to the Forecast?
# Probability Forecasts in Kalshi Weather Markets

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## Author Note

This draft extends the empirical capstone through November 4, 2026. Version 6 adds the estimated six-source public forecast and its Kalshi blend, uncertainty intervals and training-window sensitivities. Version 7 clarifies the data construction, the prospective liquidity evidence and the proposed mechanism. The new text and results are prepared for author review. Bulletin response, liquidity mechanisms and other unfinished tests remain planned.

*Use of AI tools.* I used AI assistants (Anthropic's Claude and OpenAI's Codex) to help write and check data-processing and estimation code, validate data, and draft and edit the prose of this report from my own written contributions. The research question, test predictions, final specification choices, and interpretations are my own, and the new Version 7 material is awaiting my review.

**Abstract**

Do prediction-market prices add to public weather forecasts? I compare six-bin Kalshi probabilities with NOAA's National Blend of Models (NBM) and a combination of six public weather sources at the original forecast cutoffs. In 139 chronologically held-out New York events, Kalshi's mean Brier loss is 0.1001 against 0.1171 for NBM, a 14.5% reduction. Exploratory comparisons in Los Angeles, Miami and Phoenix point the same way. Later NBM revisions move toward earlier market probabilities, while station corrections learned from past outcomes remove part of the original gap in Los Angeles and Miami. The six-source extension uses a Gaussian public model trained on 60 earlier eligible events and a past-trained probability blend with Kalshi. Across 1,184 common evaluation events, public aggregation improves on original NBM in New York, Los Angeles and Miami under the primary Brier intervals. Kalshi retains lower loss and adds predictive value to the public model in New York and Los Angeles; Miami and Phoenix are inconclusive. Training-window sensitivity is material, with Los Angeles giving the most consistent evidence across the 40-event, 60-event and expanding variants. These findings support incremental predictive value relative to the specified public model. They leave the information source unresolved and coexist with the earlier evidence of overpriced longshots. Separate-city inference uses paired blocks of consecutive event-days.

*Keywords:* prediction markets, weather forecasts, probabilistic forecasting, information aggregation, Kalshi

## Introduction

In an economy that increasingly trades on information, markets that price information directly deserve careful evaluation. Prediction markets rest on an old idea: a market can gather information scattered across many individuals into a single price (Wolfers & Zitzewitz, 2004). Weather is a demanding test of that idea. A government forecasting system already blends dozens of models into a calibrated probability forecast that anyone can consult for free. What could a few traders know that this forecast does not? They may simply repackage the same public information and add their own judgment. Even so, a price that aggregates many such judgments might forecast better than any single source.

I study Kalshi's daily-high-temperature markets for this reason. Few outcomes a market can price are modeled, observed, and archived as often and as completely as weather, so traders and researchers alike have ample public information to measure the market against. Each contract settles daily on an official station reading, which yields hundreds of resolved forecasts per city. Everyone experiences weather, and participants plausibly range from casual speculators to commodity traders hedging weather exposure, although the data cannot identify them. If a prediction market can add anything to a strong public forecast, weather is where that claim can be checked most cleanly.

I begin with a side-by-side comparison. At 10 p.m. on the evening before each event, I convert Kalshi prices across six temperature brackets into a probability distribution and convert percentiles from NOAA's National Blend of Models (NBM) into probabilities for the same brackets. I then score both against the same settled outcome. Accuracy alone does not show that traders know more than the benchmark, but it is the necessary first step: if Kalshi is more accurate, the question becomes why.

Kalshi has lower forecast loss in all four samples. On 139 chronologically held-out New York dates, Kalshi's mean Brier loss is 14.5% below NBM's, and a combination trained only on earlier outcomes places all its weight on Kalshi. In Los Angeles, Miami, and Phoenix (1,235 events), Kalshi again has lower loss in the full retrospective samples and on the later evaluation

events of a trained combination, where mixing in NBM does not improve on Kalshi alone; these city comparisons are exploratory. Kalshi also beats a stricter benchmark: the first NBM forecast archived after the cutoff, typically two to five hours later. That run has information the cutoff price could not have, yet it still has higher loss in every city, although the margin cannot be distinguished from zero in Phoenix's smaller evaluation subset. What surprised me most is how those revisions behave. They tend to move in the direction of the earlier Kalshi forecast, but they do not reliably end up any closer to it, and they clearly improve NBM's own accuracy only in Phoenix. Finally, a correction that shifts NBM by the station error it made on earlier, already published outcomes narrows the gap substantially in Los Angeles and Miami, but Kalshi still has lower loss than the corrected benchmark in every city.

Why is this market more accurate than a well-calibrated public benchmark? One possibility is that aggregating many traders' judgments produces a better forecast even if no individual trader holds superior information. Another is that the comparison is not entirely even: the market may hold a structural advantage over the benchmark. The clearest candidate is the target itself. Kalshi settles on a single station thermometer, while NBM forecasts a 2.5-kilometer grid cell, so a market that learns a station's persistent quirks starts with an advantage that has nothing to do with private information. Other structural advantages may exist that this paper has not yet identified. The descriptive evidence points first to a *location bias*: over the sample, NBM's median at the settlement station ran about 1.8°F too cold in Los Angeles and 1.0°F too warm in New York, and the market sat on the correcting side. A bias that persists is learnable from past outcomes, so I test a correction trained only on earlier data (P3). It accounts for part of the gap in Los Angeles and Miami, not all of it, and none in New York or Phoenix. Throughout, I keep three claims separate: *accuracy* (lower forecast loss), *incremental predictive content* (improvement over a specified alternative), and *information source* (why the improvement occurs). This paper addresses the first two for one public benchmark. It does not identify private signals.

Two recent studies address the same operational question. Crosier (2026) compares market-implied mean temperatures with public forecasts across seven cities and reports that later

NBM values move toward earlier market prices. Brown (2026) scores full ForecastEx temperature distributions against ensembles recalibrated for location. My contribution is (a) a matched-cutoff comparison of complete six-bin Kalshi distributions against NOAA's own archived probabilistic NBM, scored with proper rules on the contract's settled outcome; (b) out-of-sample combination tests that separate standalone accuracy from complementary information; (c) an audit of timing, the physical target window, and archive availability; and (d) a sequence of tests designed to tell explanations apart, including whether NBM revisions converge toward the market and whether the advantage survives a past-trained station correction. The aim is not only to show that the market wins, but to ask what it is winning on.

## Literature Review

### Information Aggregation and the Meaning of Prices

The literature on information aggregation sets this paper's structure. Markets gather scattered judgments into prices, and prediction markets apply that property to information itself (Wolfers & Zitzewitz, 2004). Prices cannot be fully informative, however. Grossman and Stiglitz (1980) show that when acquiring information is costly, prices that revealed everything would leave no reward for gathering it, so some mispricing must remain to compensate the traders who do. A market can therefore add information while remaining imperfect. That imperfection may also act as a correcting mechanism: mispriced probabilities create an incentive for better-informed traders to trade against them. This is a hypothesis about why markets might outperform, not something these data can show; without trader-level data, I cannot model who those traders are.

Reading prices as probabilities requires care. In my view, a contract price combines two components: the trader's judgment about the outcome, and the incentives of trading for profit, such as risk attitudes, position limits, and the wish not to reveal what one knows. Manski (2006) formalizes this concern: an equilibrium price need not equal the average belief of traders without strong assumptions. I therefore treat normalized midpoints as *market-implied probabilities*, forecasts to be evaluated rather than literal beliefs.

**Markets Versus External Benchmarks**

Prediction markets have long been compared with outside forecasts, which motivates comparing Kalshi with a scientific benchmark. The more demanding question is whether a market adds *new* information beyond the benchmark, not merely whether it scores better, and that is a central aim of this study. The benchmark matters. Erikson and Wlezien (2012) find that election markets add little beyond polls once the polls are properly transformed, so beating a raw benchmark is weaker evidence than beating a well-constructed one. This is why the project plans a past-trained station correction of NBM. Mayew and Zhu (2026) draw the same line in Kalshi's earnings-call mention contracts, which are highly predictive on their own but add only modest information beyond public disclosures: the distinction between standalone and incremental value on which my combination tests are built.

Trading incentives can also bend market probabilities in systematic ways. Page and Clemen (2013) document favorite–longshot bias in prediction markets: unlikely outcomes tend to be overpriced and likely ones underpriced. A market can therefore beat a benchmark on average while overstating particular unlikely outcomes, which is one reason the comparison with a calibrated public forecast is not perfectly like-for-like. Test T4 shows this pattern in New York.

**Weather Markets**

The weather-market literature is developing quickly. Two recent studies examine the question this paper asks and define what remains open. Crosier (2026) compares market-implied *mean* temperatures with public weather products across seven cities, reporting that later NBM forecasts tend to move toward earlier market prices and that a public-only forecast combination narrows the market's advantage. Its station-bulletin forecasts and timing differ from this project's percentile reconstruction. Brown (2026) evaluates full ForecastEx temperature distributions against ensembles recalibrated for location and dispersion. Neither distributional scoring nor location correction is therefore new by itself. What this paper adds is a full six-bin Kalshi

distribution compared with NOAA's own probabilistic NBM, out-of-sample combination tests, and a sequence of tests aimed at the source of the advantage.

**Public-Information Processing and Microstructure**

How prices absorb public news bears directly on the mechanism question. Angelini and De Angelis (2026), using minute-level Kalshi basketball quotes, find that prices move in the right direction after public information but adjust incompletely on impact, with the missing adjustment predicting later drift when liquidity is low. My revision test finds a related pattern from the other side: later NBM forecasts move toward the earlier Kalshi forecast without converging to it. Whether the two reflect a common mechanism is an open question this paper can begin to explore.

Market quality matters for a second reason. If only a few traders move a market, its price aggregates little dispersed information. Microstructure theory explains why bid–ask spreads exist when some traders are better informed (Glosten & Milgrom, 1985) and how the information content of trades can be measured through their lasting effect on prices (Hasbrouck, 1991). Spreads and volume are not direct measures of informed trading, and historical order-book depth is not available for these markets, so this paper examines market quality through spreads, volume, and public trades.

**Forecast Evaluation**

Scoring the forecasts properly, and designing a test that could speak to superior information, was among the most demanding parts of this project. Strictly proper scoring rules reward a forecaster for reporting the true predictive distribution, which allows whole distributions to be evaluated rather than only their means (Gneiting & Raftery, 2007). The Brier score separates into reliability (whether stated probabilities match realized frequencies), resolution (whether forecasts distinguish the days on which an outcome is more or less likely), and uncertainty (how hard the outcomes are to predict at all) (Murphy, 1973). This decomposition can show where Kalshi's advantage comes from: better resolution points to information, better reliability to calibration. It is a planned test (S3). Differences in loss between two forecasts must

be compared while allowing for dependence across consecutive days (Diebold & Mariano, 1995). Cross-calibration asks whether one forecast remains calibrated once the other is known (Strähl & Ziegel, 2017), and any evaluation focused on extreme outcomes must avoid selecting cases by what actually happened (Lerch et al., 2017).

## Data

### Contracts and Outcome

Kalshi's KXHIGH series is a natural forecast comparison: each event is itself a probability forecast of one station's daily high. It lists six mutually exclusive and exhaustive bins: four interior bins two degrees Fahrenheit wide and two open-ended bins. The outcome for event $i$ is the one-hot vector $y_i$ identifying the bin Kalshi settled as YES, and both forecasts are scored against it. In the NYC held-out sample, 125 events settled on the NWS Climatological Report and 14 on The Weather Company after Kalshi's source change. Across all scored events, the official climate-report (CLI) high falls in the settled bin for 1,427 of 1,435 events; the remaining eight lack a CLI high, and retrospective ACIS values place all eight in the settled bin. One resolved event-day is one forecast case; its six bins are dependent components, not six independent outcomes. The settlement stations are Central Park (New York), Los Angeles International Airport (KLAX), Miami International Airport (KMIA), and Phoenix Sky Harbor (KPHX).

### Market Probabilities

Kalshi probabilities use each bin's latest YES bid–ask midpoint at or before the cutoff, subject to a maximum quote age of 180 minutes, since a stale quote is not a current forecast. Because each price embeds part of the spread, and longshot bins tend to be overpriced, the six midpoints sum to more than one; the sum averages 1.034 in New York. Dividing each midpoint by the sum preserves the market's ranking of bins and removes the 3.4-percentage-point overround on the typical event. Two prospective order-book snapshots (September 22–23, 2026, outside the sample) show the same overround at execution prices: buying all six YES contracts at

the best asks cost $1.08 and $1.06 for a set paying exactly $1. NYC uses hourly Kalshi candles; the city extension uses one-minute candles.

### Choosing a Public Benchmark

Public weather forecasts differ in how much work it takes to turn them into a bin probability. Consumer-facing products, such as the forecast high a person sees on weather.gov or a phone app, give a single number. Free professional guidance, such as NBM percentiles and station Model Output Statistics (MOS), is public but must be converted into probabilities. Raw ensembles, such as NOAA's GEFS, the ECMWF ensemble, and the Canadian GEPS, require substantial processing. I classify benchmarks this way so that the study does not simply ask whether a market beats the most complex model available, and so that it does not assume every trader is an informed trader. The primary benchmark is NOAA's probabilistic NBM, a calibrated blend that anyone can download. Comparisons with ensembles and public-only combinations are planned tests.

### Public Benchmark Construction

The benchmark uses NBM QMD maximum-temperature percentiles at the grid point nearest each station; the median station–grid distance is 0.37–0.85 km. Percentiles are converted to Fahrenheit, crossed quantiles are repaired by isotonic regression, and a cumulative distribution is interpolated linearly with bounded tails and differenced at half-degree bin boundaries (Appendix A). QMD cycles run at 00, 06, 12, and 18 UTC, unlike the hourly Core product (National Centers for Environmental Prediction [NCEP], 2026).

### Timing and Physical Target

Each event carries six separate clocks: NBM cycle time, archive time, market open, forecast cutoff, physical target window, and settlement. For New York the cutoff is fixed at 02:00 UTC on the target date. Every one of the 139 held-out dates falls in daylight saving time, so the cutoff is 10 p.m. EDT throughout evaluation; only the first 28 training dates, before daylight

saving began on March 8, 2026, had a 9 p.m. EST cutoff. The city extension uses 10 p.m. local time. These are forecast cutoffs, not market closing times.

The NBM vintage is the latest cycle whose archived file exists by the cutoff. NOAA disseminates NBM in real time and also copies it to a public cloud archive. The archive's Last-Modified timestamp records when the file was last written there, not when the forecast first reached the public or any trader, so it is an availability proxy. If the true release preceded the archive copy, traders may sometimes have had a slightly fresher run than the one used here. Two checks bound this concern. Requiring files to be archived at least 15 or 60 minutes before the cutoff leaves Miami's gap, the city with the shortest median archive lead (43 minutes), at 0.0132 and 0.0126 (Table A7). And a later NBM run archived after the cutoff still loses to the cutoff market price (test S1).

NBM's maximum-temperature field covers 12:00 UTC on the target date through 06:00 UTC the following date, whereas Kalshi settles the station's local-day high. In New York, observed station temperatures give the same maximum in the two windows on 186 of 200 dates and the same contract bin on 194 (Table A5). In a hash-selected audit of 35 city events, the official high was in the settled bin and its reported time inside the NBM window in all 35 cases, under both readings of the report's clock. One Miami date with an early high remains in the sample. The cutoff precedes the NBM window by about 9–10 hours in New York and Miami and 6–7 hours in Los Angeles and Phoenix. Together with the different evaluation protocols, this is why the four samples are reported separately rather than pooled.

**Samples**

Table 1 summarizes sample construction. The NYC panel contains 200 events from February 9 through August 27, 2026; the first 61 serve only as initial training, and 139 chronological forecasts from April 11 through August 27 form the evaluation sample. The city extension downloaded 1,368 events; 53 fail structural checks and 80 fail the quote-freshness rule,

leaving 1,235 paired events. Every selected city NBM baseline was archived before its local cutoff.

**Table 1**
*Sample Construction and Forecast Timing*

| | New York | Los Angeles | Miami | Phoenix |
|---|---|---|---|---|
| Evaluation design | Held-out | Retrospective | Retrospective | Retrospective |
| First / last target date | 04-11 / 08-27 | 2025-01-05 / 08-13 | 2024-12-29 / 08-12 | 02-04 / 08-12 |
| Events selected / scored | 200 / 139 | 586 / 543 | 592 / 505 | 190 / 187 |
| Cutoff (local) | 10 p.m. EDT | 10 p.m. | 10 p.m. | 10 p.m. |
| Kalshi candle interval | 1 hour | 1 minute | 1 minute | 1 minute |
| NBM cycles used (00/12/18 UTC) | 1/76/62 | 2/3/538 | 2/89/414 | 1/3/183 |
| Median NBM cycle age at cutoff (h) | 14 | 11 | 8 | 11 |
| Median archive lead before cutoff (min) | 401 | 212 | 43 | 180 |
| Median station–grid distance (km) | — | 0.85 | 0.59 | 0.37 |

*Note.* Dates without a year are 2026. New York "selected" is the full 200-event panel; 61 early events serve only as initial training. City exclusions combine 53 structural failures and 80 stale-quote failures. Cycle age is cutoff minus nominal cycle time; archive lead is cutoff minus the archive Last-Modified time.

## Revision Data

The revision tests compare the cutoff NBM with the first NBM run archived after the cutoff and before the target window begins. Both are rebuilt from all available interior percentiles at the same grid point, so a difference between them reflects a forecast revision rather than a change in construction. Of 1,435 events, 1,423 have an eligible later run (New York 198, Los Angeles 539, Miami 501, Phoenix 185); the 12 without one reflect timing coverage, not performance. The median delay from cutoff to the revision is 2.3–2.5 hours, and 5.2 hours in Miami.

## Validation

Every dataset was validated before any score was computed, and each construction was written to a new file rather than overwriting earlier ones. For the revision tests, a separate script independently rebuilt all 2,858 NBM curves, agreeing with the original to within $8.4 \times 10^{-14}$ per

bin, and rehashed all 98,984 raw files against their recorded checksums. Only after this check passed was the test specification fixed and estimated. Appendix A describes the procedures.

**Additional Data for Planned Tests**

The project also holds 5.5 million public Kalshi trades, evening station observations (ASOS), station model guidance (MOS/LAMP), ensemble forecasts (GEFS, ECMWF, and Canadian GEPS), and, acquired since the October 4 draft, NOAA's gridded NDFD and HRRR forecasts, the ECMWF open-data feed, and the local NWS forecast bulletins. These feed the planned tests and are described where each test is introduced. The TIGGE ensemble archive could not be accessed until roughly 30–34 hours after the cutoff, so it serves only as a retrospective comparison. The Canadian GEPS ensemble is set aside from the planned public benchmarks because its historical release times could not be verified without paying for the evidence.

**Descriptive Statistics**

Table 2 describes the 139 New York evaluation events. Three facts matter for interpreting the comparison. First, the markets were active: over the roughly 12 hours each market was open before the cutoff, a median of 17,108 contracts traded across the six bins, so the cutoff price is not a stale artifact of a few trades. Second, quotes were tight and current: the mean bid–ask spread at cutoff was 1.42 cents per $1 contract, and in the median event every selected quote was current at the cutoff. City markets were thinner, with mean spreads of 1.8–2.7 cents and older quotes (Table A1), so their probabilities are somewhat noisier. Third, the two forecasts are about equally spread out. Normalized entropy, a 0-to-1 measure of how spread out a distribution is, averages 0.69 for Kalshi and 0.70 for NBM, so Kalshi does not win simply by being bolder or more cautious. NBM does place more probability in the open-ended bins (0.31 versus 0.25).

**Table 2**
*New York City Held-Out Sample: Market and Benchmark Inputs*

| Variable | Mean | SD | Min | Median | Max |
|---|---|---|---|---|---|
| Settlement high (°F) | 77.9 | 10.4 | 50 | 81 | 100 |
| Hours open before cutoff | 12.0 | 0.3 | 8.8 | 12.0 | 12.0 |
| Contracts traded, open to cutoff | 18,359 | 6,684 | 6,801 | 17,108 | 38,558 |
| Premium turnover, open to cutoff ($) | 4,191 | 1,719 | 1,417 | 3,896 | 9,888 |
| Open interest at cutoff | 14,217 | 4,975 | 5,413 | 13,477 | 29,152 |
| Mean bin spread at cutoff (cents) | 1.42 | 0.38 | 1.00 | 1.33 | 3.17 |
| Widest bin spread at cutoff (cents) | 2.51 | 1.19 | 1.00 | 2.00 | 8.00 |
| Oldest selected quote (minutes) | 3.5 | 14.0 | 0 | 0 | 60 |
| Raw sum of six midpoints | 1.034 | 0.027 | 0.960 | 1.035 | 1.105 |
| NBM P90 − P10 (°F) | 5.99 | 1.70 | 3.24 | 5.76 | 12.24 |
| Normalized entropy, Kalshi | 0.692 | 0.118 | 0.209 | 0.707 | 0.910 |
| Normalized entropy, NBM | 0.704 | 0.163 | 0.108 | 0.718 | 0.990 |
| Open-bin probability, Kalshi | 0.250 | 0.226 | 0.009 | 0.162 | 0.926 |
| Open-bin probability, NBM | 0.306 | 0.220 | 0.000 | 0.240 | 0.952 |

*Note.* $n$ = 139 event-days. Trading measures sum all six bins. Normalized entropy is six-bin Shannon entropy divided by ln 6 (0 = all mass in one bin, 1 = uniform). Open-bin probability is the mass on the two open-ended bins.

Table 3 compares the shape of the two distributions. The difference lies in *where* each forecast places its probability, not *how much* it concentrates. Kalshi places more probability on the bin that wins in every city (0.36–0.40, against 0.21–0.35 for NBM), and its most likely bin wins 51–55% of the time, against 21–48% for NBM. In Los Angeles NBM is the sharper forecast but the worse one: its top bin carries 0.53 on average against Kalshi's 0.45, yet it places only 0.21 on the realized bin. NBM is confidently misplaced. Location error, the realized bin minus each forecast's expected bin, points the same way: NBM is off by −0.40 to +0.90 bins, while Kalshi is within 0.16 bins in every city. Kalshi's advantage appears to come from centering its forecast in the right place rather than from being more or less confident.

**Table 3**
*Shape and Location of the Two Forecast Distributions*

| | New York | Los Angeles | Miami | Phoenix |
|---|---|---|---|---|
| Events | 139 | 543 | 505 | 187 |
| Largest bin probability, Kalshi / NBM | 0.47 / 0.44 | 0.45 / 0.53 | 0.49 / 0.47 | 0.50 / 0.46 |
| Probability on realized bin, Kalshi / NBM | 0.38 / 0.32 | 0.36 / 0.21 | 0.39 / 0.35 | 0.40 / 0.34 |
| Most likely bin wins, Kalshi / NBM | 0.53 / 0.39 | 0.51 / 0.21 | 0.55 / 0.44 | 0.55 / 0.48 |
| Open-bin probability, Kalshi / NBM | 0.25 / 0.31 | 0.17 / 0.20 | 0.08 / 0.07 | 0.06 / 0.07 |
| Share of outcomes in an open bin | 0.30 | 0.15 | 0.08 | 0.04 |
| Total variation distance, Kalshi vs. NBM | 0.26 | 0.51 | 0.23 | 0.22 |
| *Location error: realized minus expected bin index (+ = forecast too cold)* | | | | |
| NBM | −0.40 | +0.90 | +0.37 | −0.13 |
| Kalshi | −0.15 | +0.02 | +0.16 | −0.03 |

*Note.* All entries are event means. Expected bin index is $\sum_j j\, p_{ij}$ over ordered bins $j = 1, \ldots, 6$; interior bins are 2°F wide, so one index unit is about 2°F. Total variation distance is $\frac{1}{2} \sum_j |p_{ij}^K - p_{ij}^N|$ (0 = identical, 1 = no overlap). Full-sample descriptive statistics, not estimated corrections.

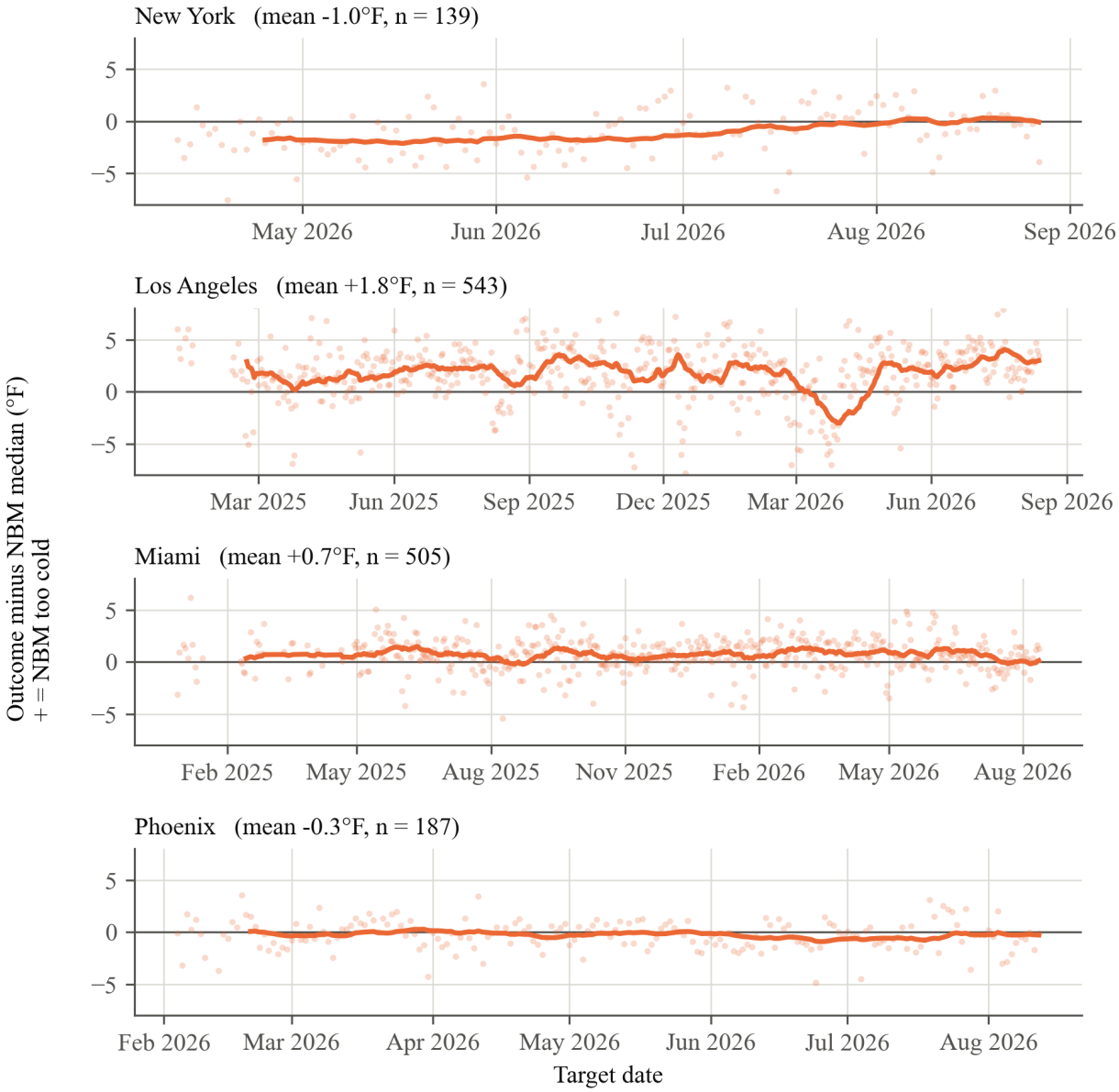
## What Is the NBM Location Bias?

Think of a thermometer that always reads two degrees warm. It may be perfectly consistent, but every reading is off in the same direction. A forecast can behave the same way: its *spread*, how uncertain it is, may be right, while its *location*, where it is centered, sits persistently off the outcome. That persistent offset is a location bias. I measure it as the official daily high minus the NBM median at the station's grid point, using exact official highs in all four samples.

Figure 1 shows the result. NBM's median ran 1.8°F too cold in Los Angeles (95% interval 1.3–2.2), 0.7°F too cold in Miami (0.5–0.9), 1.0°F too *warm* in New York (0.5–1.5), and 0.3°F too warm in Phoenix (0.04–0.6). Two features make the bias matter. It is large relative to the contracts: with 2°F bins, a 1.8°F shift moves most of NBM's probability one bin away from the outcome, which is what the Los Angeles rows of Table 3 show. And it is persistent: the 30-event rolling mean in Los Angeles stays above zero for most of 20 months, while New York's bias holds through June and then fades toward zero in July and August, the same months in which Kalshi's New York advantage flattens (Figure 3).

**Figure 1**

*NBM Median Forecast Error at Each Settlement Station*




*Note.* Outcome minus the NBM median at the station's nearest grid point, in °F; positive values mean NBM was too cold. Dots are events; the line is a 30-event rolling mean. New York uses the reported high on the 139 held-out dates. The cities use the official daily high on every event (Los Angeles 543, Miami 505, Phoenix 187); eight events without a climate-report high use retrospective ACIS values. Horizontal date scales differ by panel.

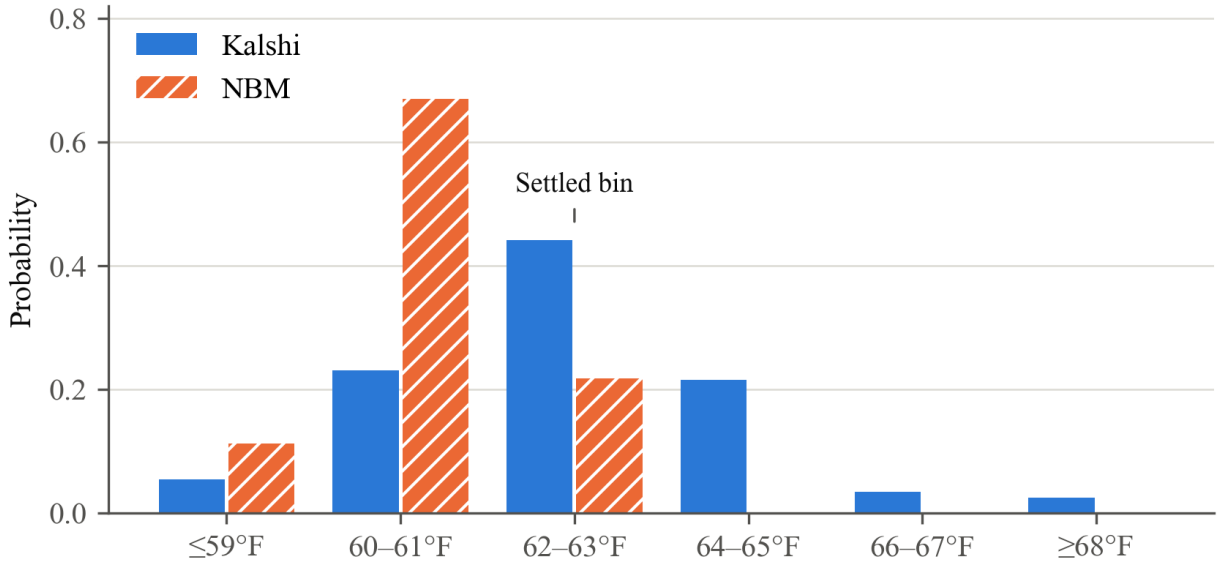
Why would an official forecast be off-center at one station? These data cannot say, but several public, physical explanations are candidates, and none requires traders to know anything private.

- **Grid versus point.** NBM forecasts a 2.5-kilometer grid cell, while the contract settles on one thermometer. Along the coast at Los Angeles International Airport, where marine air and sea breezes dominate, a station and its grid cell can differ systematically even 0.85 kilometers apart.
- **Station siting.** Each official instrument has its own exposure. The Central Park station sits in a large vegetated park, and a station reading cooler than its grid cell would produce the warm NBM error seen in New York. This is a hypothesis, not a verified fact.
- **Target window.** NBM's window and the contract's settlement day usually agree (Table A5), but not always.
- **Seasonal and version change.** The bias varies with season, and NBM version 5.0 was implemented on May 5, 2026, within the New York sample.

Whatever its source, a persistent bias is *learnable from past public data*: anyone who tracks a station's record against NBM can shift the forecast. Kalshi's distribution sits on the correcting side in every city. In the typical Los Angeles event (Figure 2), NBM places 0.67 on the 60–61°F bin, while Kalshi centers one bin warmer, where the high settled. If most of Kalshi's advantage comes from this correction, the market adds value by pricing a predictable public bias. That is useful to a forecaster, but it is not evidence of information beyond the public record. It also fits the possibility that the comparison is not entirely even: a market settling on a station competes with a forecast built for a grid cell.

**Figure 2**

*An Illustrative Los Angeles Event (April 2, 2025)*




*Note.* Six-bin probabilities at the 10 p.m. cutoff. The event was chosen, from all Los Angeles events, as the one whose NBM and Kalshi location errors are closest to the city averages in Table 3 (NBM +0.90 bins, Kalshi +0.02). Brier loss is 0.069 for Kalshi and 0.179 for NBM.

The averages above use the full sample, including outcomes a forecaster would not yet have seen, so they describe the bias without correcting for it. Test P3 (Step 7) corrects NBM using only outcomes published before each cutoff. The full-sample bias is not the same as a bias a forecaster could have learned in time: in Los Angeles and Miami the past-trained correction removes most of the gap, but in New York the error learned from earlier outcomes had the opposite sign to the in-sample error in Figure 1, and the correction does not help. Figure 1 should therefore be read as a description of the sample, not as a correction available at the time.

## Model and Results

### What Is Being Scored?

For event $i$ with six bins $j$, the multicategory Brier score of forecast $p$ is

$$B_i(p) = \frac{1}{6}\sum_{j=1}^{6}\left(p_{ij} - y_{ij}\right)^2. \tag{1}$$

Lower loss is better. The score evaluates all assigned probabilities, not just whether the most likely bin wins. The normalized ranked probability score (RPS; Appendix A) compares cumulative probabilities and so penalizes a miss by one bin less than a miss by four. The loss difference on each event is $d_i = B_i(p^N) - B_i(p^K)$, positive when Kalshi does better. Inference uses a circular moving-block bootstrap with seven consecutive event-days and 10,000 replications; a heteroskedasticity-and-autocorrelation-consistent (HAC, seven-lag) test in the spirit of Diebold and Mariano (1995) is the secondary check. The tests build on one another: each answers a question the previous one left open. Table 4 lists them in that order.

**Table 4**
*Completed Tests*

| Test | Question | Result | Where |
|---|---|---|---|
| T1 Paired scores (NYC) | Which forecast has lower held-out loss? | Kalshi; Brier gap 0.0170, 95% CI excludes zero | Table 5, Fig. 3 |
| T2 Chronological combination | Does an interior Kalshi–NBM mix beat both? | Weight on Kalshi is 1 on all 139 dates | Fig. 4 |
| T3 NBM precision | Does sparse percentile use explain T1? | No; gap 0.0171 with all interior percentiles | Text |
| T6 Target agreement | Do the two physical windows share a maximum? | 93% same high, 97% same bin | Table A5 |
| T4 Calibration and tails | Are both forecasts reliable by probability level? | Similar overall; Kalshi overprices ≤5% bins | Fig. 5 |
| T7 City replication | Does the direction hold elsewhere? | Kalshi lower loss in LA, Miami, Phoenix (exploratory) | Table 7, Fig. 6 |
| T2-city Combination | Does a trained mix beat Kalshi in the cities? | No; intervals include zero (exploratory) | Table 8 |
| T5 Market quality | Is the gap larger on active, tightly quoted days? | Not detected; gap smaller when NBM is wide | Table 6 |
| P4 First NBM revision | Does later NBM move toward Kalshi and improve? | Moves toward Kalshi; no clear convergence; Brier gain in Phoenix only (exploratory) | Table 9 |
| S1 Fresh NBM | Does Kalshi still beat a later NBM run? | Yes in all four cities; Phoenix subset inconclusive (exploratory) | Table 9 |
| P3 Station correction | Does a past-trained location correction close the gap? | Narrows LA and Miami; Kalshi still lower loss in all four full samples (exploratory) | Table 10 |
| P4/S1 corrected | Do the revision results survive the correction? | Alignment persists; Kalshi still ahead of later corrected NBM except Miami (exploratory) | Table 10 |

*Note.* Every test uses the fixed construction; none changes the cutoff, sample, or outcome after scores were observed. Tests are listed in the order discussed.

### Step 1: Does Kalshi Beat NBM? (T1–T3, New York)

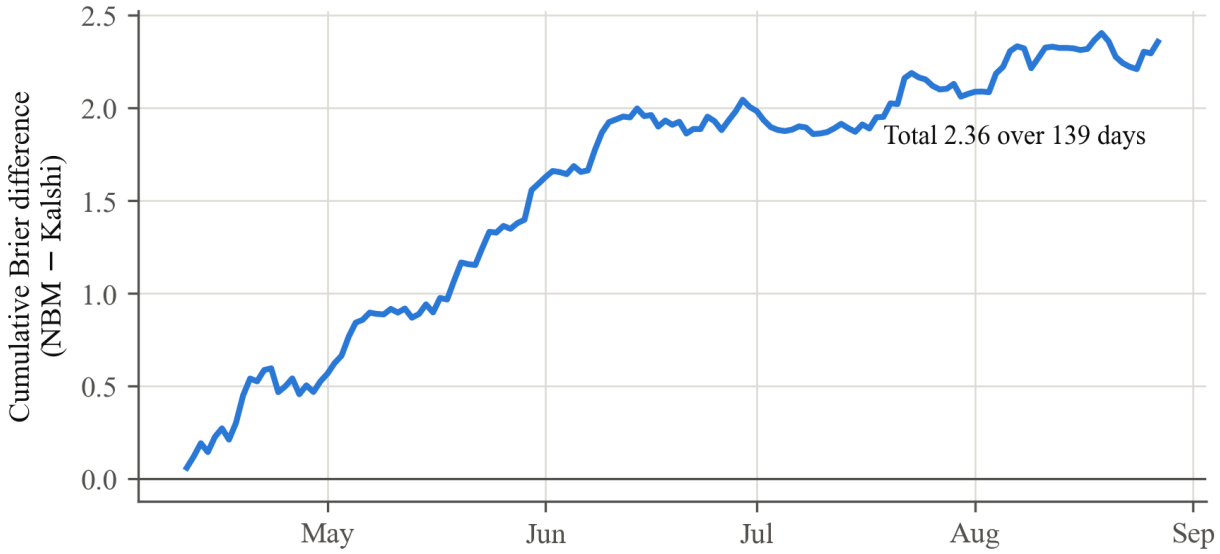
**T1. Paired held-out comparison.** On 139 chronologically held-out New York events, Kalshi's mean Brier loss is 0.1001 against NBM's 0.1171, a gap of 0.0170 [0.0078, 0.0264] and a 14.5% reduction (Table 5). Kalshi has the lower loss on 83 of 139 days, and the normalized RPS gap points the same way. The advantage accumulated mostly from April through June and was nearly flat in July, when NBM's New York bias had faded (Figure 3; Table A4). The monthly pattern is descriptive; no trend test was prespecified.

**Table 5**
*New York City Held-Out Forecast Performance*

| | Brier | Normalized RPS |
|---|---|---|
| Mean loss, Kalshi | 0.100104 | 0.078968 |
| Mean loss, NBM | 0.117060 | 0.112088 |
| Difference (NBM − Kalshi) | 0.016956 | 0.033120 |
| 95% block-bootstrap CI | [0.007826, 0.026430] | [0.019297, 0.047496] |
| Relative reduction | 14.48% | 29.55% |
| Block-bootstrap $p$ / HAC $p$ (two-sided) | 0.0002 / 0.0003 | 0.0001 / — |
| Days Kalshi has lower loss | 83 of 139 | 90 of 139 |

*Note.* $n$ = 139 event-days, April 11–August 27, 2026. Positive differences favor Kalshi. The trained combination equals Kalshi on every date, so its losses equal Kalshi's. The HAC test was run for the primary Brier score only.

**Figure 3**
*Cumulative Kalshi Advantage Over the New York Held-Out Sample*




*Note.* Running sum of the daily Brier difference $d_i$ (NBM − Kalshi). A rising line means Kalshi is outscoring NBM; a flat line means the two are even. Dividing the final value by 139 gives the mean difference in Table 5.

**T2. Chronological combination.** The incremental-information test combines the two forecasts:

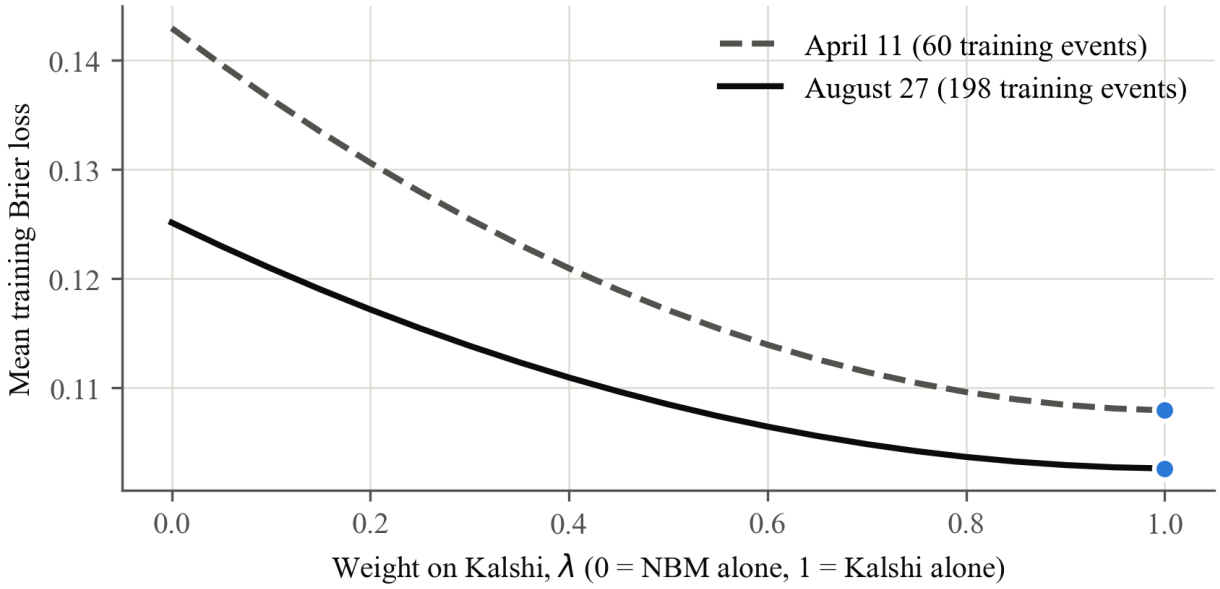
$$p_{ij}^{C} = \lambda_i p_{ij}^{K} + (1 - \lambda_i) p_{ij}^{N}, \qquad 0 \le \lambda_i \le 1. \tag{2}$$

For each target date, $\lambda_i$ minimizes past mean Brier loss on a grid from 0 to 1 in steps of 0.05, over an expanding window of at least 60 events whose outcomes were available before that target's

cutoff. The procedure selects $\lambda_i = 1$ on every held-out date, and training loss is still falling at that upper bound (Figure 4; Table A2). This corner solution means something specific: within a simple weighted average, NBM adds nothing once Kalshi is known. It does not show that NBM contains no useful information. Because the constraint $\lambda \leq 1$ binds, this test cannot detect complementarity; the completed public combination in Step 8 broadens the comparison, while unrestricted encompassing weights remain a separate S8 extension.

**Figure 4**

*Training Loss by Weight on Kalshi*




*Note.* Mean Brier loss over each date's training window for every candidate weight. The selected weight (dot) is at the upper bound on both dates.

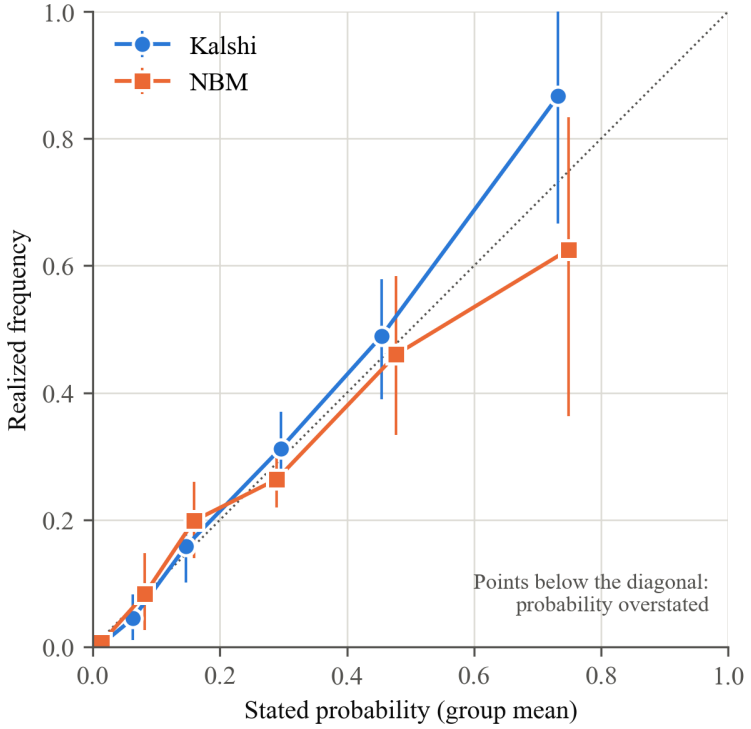
**T3. NBM precision.** Rebuilding every NBM forecast from all available interior percentiles (99 on 86 dates, 19 on 114) rather than the 13 or 15 most vintages expose leaves the gap at 0.0171 [0.0081, 0.0265], with no selected weight changed. The advantage does not appear to be an artifact of how the benchmark was reconstructed. These three tests establish the result for one city and one sample. A single city cannot show that the pattern generalizes, which motivates the checks and replications that follow.

**Step 2: Is It a Measurement Artifact? (T6, T4)**

**T6. Physical-target agreement.** Across 200 New York dates, the contract and NBM windows produce the same observed maximum on 93% of dates and the same bin on 97% (Cohen's $\kappa = 0.96$); on average the NBM window's maximum was 0.29°F cooler (Table A5). Removing the one known Miami mismatch moves Miami's gap only from 0.01160 to 0.01162. Window mismatch appears too rare to explain an average bias of 1–2°F. This check is a precondition for the comparison rather than a finding about either forecast.

**T4. Calibration and tails.** Figure 5 groups the 834 held-out bin forecasts by stated probability. Overall, the two forecasts are similarly calibrated: the expected calibration error is 1.75 points for Kalshi and 1.74 for NBM. The difference lies at the extremes. Kalshi's 345 bins priced at 5% or less averaged 1.78% but won only once (0.29%), and its 15 most confident bins won 87% of the time against a stated 74% (Table A3). This is the favorite–longshot pattern (Page & Clemen, 2013). Kalshi therefore wins on average *despite* overpricing unlikely outcomes. For a consumer, a 3% Kalshi price is not a 3% chance: in this sample such bins won far less often, so buying them carried a negative expected return, although they may still appeal to a risk-seeking buyer who values a small chance of a large payoff. Because this mispricing raises Kalshi's Brier loss, correcting it would more likely widen the gap than close it. Planned tests will recalibrate both forecasts before asking where the advantage comes from.

**Figure 5**
*New York City Reliability Diagram*




*Note.* 834 bin forecasts from 139 event-days, grouped at 0–5, 5–10, 10–20, 20–40, 40–60, and 60–100%. Vertical bars are 95% block-bootstrap intervals for the realized frequency. Bins within an event are dependent; with a simultaneous correction across groups, no single gap is distinguishable from zero.

### Step 3: Does It Replicate? (T7)

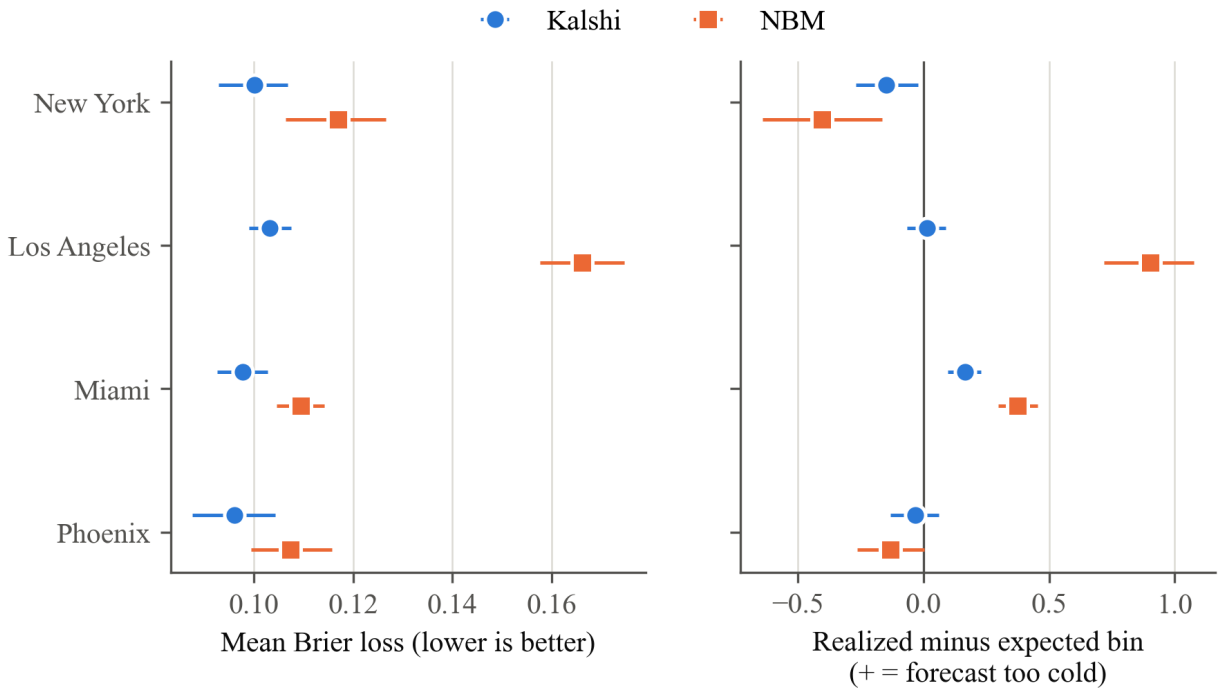
**T7. City replication.** Applying the same construction at 10 p.m. local time, Kalshi has lower mean Brier loss in Los Angeles (0.0628 [0.0530, 0.0725]), Miami (0.0116 [0.0073, 0.0159]), and Phoenix (0.0113 [0.0047, 0.0176]) (Table 7). These comparisons are retrospective and exploratory. Kalshi's loss is nearly constant across cities (0.096–0.103), while NBM's ranges from 0.107 to 0.166, so the variation in the advantage comes from the benchmark (Figure 6). The city with NBM's largest location bias, Los Angeles, has by far the largest gap. The gap also varies by season, and requiring the NBM file to be archived well before the cutoff does not reduce Miami's gap (Table A7).

**Table 7**
*Provisional City Comparisons at 10 p.m. Local Time*

| City | *n* | Kalshi | NBM | Difference | 95% CI | Kalshi wins |
|---|---|---|---|---|---|---|
| *Brier* | | | | | | |
| Los Angeles | 543 | 0.1033 | 0.1660 | 0.0628 | [0.0530, 0.0725] | 75.5% |
| Miami | 505 | 0.0978 | 0.1094 | 0.0116 | [0.0073, 0.0159] | 59.2% |
| Phoenix | 187 | 0.0962 | 0.1075 | 0.0113 | [0.0047, 0.0176] | 61.5% |
| *Normalized RPS* | | | | | | |
| Los Angeles | 543 | 0.0861 | 0.1963 | 0.1102 | [0.0938, 0.1270] | 76.4% |
| Miami | 505 | 0.0755 | 0.0903 | 0.0148 | [0.0095, 0.0202] | 62.2% |
| Phoenix | 187 | 0.0704 | 0.0807 | 0.0103 | [0.0036, 0.0165] | 64.2% |

*Note.* Difference is NBM minus Kalshi. Seven-event-day block intervals with 4,000 replications per city. "Kalshi wins" is the share of events with lower Kalshi loss. Approximate-target retrospective comparisons with unequal histories; not pooled with New York's held-out design.

**Figure 6**
*Forecast Loss and Location Error by Sample*




*Note.* Means with 95% seven-day block-bootstrap intervals (2,000 draws, plotting only). New York is the held-out sample; the cities are retrospective. Location error is the realized bin index minus the forecast's expected bin index, as in Table 3.

### Step 4: Does NBM Add Anything to Kalshi in the Cities? (T2-city)

T7 shows that Kalshi wins in each city, and T2 showed that NBM adds nothing to Kalshi in New York. T2-city asks whether that also holds in the cities, applying the New York combination rules unchanged after 60 earlier settled outcomes. My prediction was that the mixture would not improve on Kalshi, especially in Los Angeles, and that Kalshi would receive all the weight. The mixture does not improve on Kalshi in any city (Table 8). The average weight on Kalshi is 0.996 in Los Angeles, 0.973 in Miami, and 0.959 in Phoenix, and every mixture-minus-Kalshi interval includes zero; both the mixture and Kalshi beat NBM. The prediction is mostly supported, but not literally: the selected weight was sometimes below one, so NBM occasionally received a small share. Intervals that include zero show no detectable improvement; they do not show that the two forecasts are equivalent.

**Table 8**
*Kalshi–NBM Combination in the Three Cities*

| City | Events | Mean $\lambda$ | $\lambda = 1$ share | Kalshi | NBM | Mixture − Kalshi [95% CI] |
|---|---|---|---|---|---|---|
| Los Angeles | 482 | 0.996 | 93% | 0.1019 | 0.1685 | 0.00004 [−0.00002, 0.00011] |
| Miami | 444 | 0.973 | 46% | 0.0989 | 0.1092 | 0.00011 [−0.00002, 0.00025] |
| Phoenix | 126 | 0.959 | 66% | 0.0944 | 0.1055 | 0.00030 [−0.00013, 0.00078] |

*Note.* Mean Brier loss on evaluation events after 60 earlier settled outcomes. $\lambda$ is the weight on Kalshi, selected each date from 0 to 1 in steps of 0.05 by past Brier loss. Positive differences mean the mixture is worse than Kalshi. Seven-event block-bootstrap intervals, 10,000 draws. Both Kalshi and the mixture have lower loss than NBM in every city, with intervals excluding zero.

### Step 5: Is the Gap About Market Quality? (T5)

If trading brings information into prices, Kalshi's advantage should be larger on more active, better-quoted days. Three regressions of the daily loss difference $d_i$ ask this directly, each estimated separately in the four samples:

$$\text{M0 (unconditional):} \quad d_i = a + e_i,$$

$$\text{M1 (trading activity):} \quad d_i = a + b_1 z_i^{\text{activity}} + b_2 z_i^{\text{uncertainty}} + b_3 z_i^{\text{time}} + e_i,$$

$$\text{M2 (quoted liquidity):} \quad d_i = a + c_1 z_i^{\text{spread}} + c_2 z_i^{\text{uncertainty}} + c_3 z_i^{\text{time}} + e_i.$$

Activity is log contracts traded before the cutoff, uncertainty is NBM's P90–P10 width, and each regressor is standardized within its sample. I predicted $b_1 > 0$ and $c_1 < 0$, with both effects stronger in the thinner Miami and Phoenix markets; the specification was fixed before estimation. The New York results are exploratory because an earlier joint model on the same 139 days had already been seen (Table A8).

**Table 6**
*Is Kalshi's Advantage Larger in More Active, Better-Quoted Markets?*

| | New York | Los Angeles | Miami | Phoenix |
|---|---|---|---|---|
| Events | 139 | 543 | 505 | 187 |
| *M0: unconditional advantage* | | | | |
| Mean gap | 0.0170 | 0.0628 | 0.0116 | 0.0113 |
| | [0.0078, 0.0262] | [0.0530, 0.0726] | [0.0075, 0.0159] | [0.0045, 0.0178] |
| *M1: trading activity (predicted +)* | | | | |
| Log volume | −0.0022 | −0.0075 | −0.0008 | 0.0030 |
| | [−0.0116, 0.0075] | [−0.0210, 0.0053] | [−0.0073, 0.0053] | [−0.0026, 0.0088] |
| NBM width | −0.0001 | −0.0239 | −0.0081 | 0.0016 |
| | [−0.0089, 0.0093] | [−0.0321, −0.0166] | [−0.0129, −0.0033] | [−0.0079, 0.0087] |
| Time | −0.0110 | 0.0199 | −0.0024 | 0.0015 |
| | [−0.0213, 0.0011] | [0.0057, 0.0349] | [−0.0085, 0.0043] | [−0.0042, 0.0079] |
| $R^2$ | 0.039 | 0.116 | 0.032 | 0.008 |
| *M2: quoted liquidity (predicted −)* | | | | |
| Cutoff spread | 0.0007 | 0.0030 | 0.0001 | −0.0046 |
| | [−0.0089, 0.0081] | [−0.0060, 0.0109] | [−0.0047, 0.0061] | [−0.0100, 0.0006] |
| $R^2$ | 0.038 | 0.114 | 0.032 | 0.012 |
| *Correlation of activity with time* | | | | |
| $r$ | −0.57 | 0.84 | 0.66 | −0.01 |

*Note.* Outcome is NBM minus Kalshi Brier loss; positive favors Kalshi. Coefficients are per within-sample standard deviation, with 95% seven-day block-bootstrap intervals (10,000 draws) in brackets. NBM width is the P90–P10 uncertainty control. M2's uncertainty and time coefficients are close to M1's and are omitted. New York is exploratory. Specification: `robustness_checks/MARKET_QUALITY_SEPARATE_MODELS_SPEC_2026-09-28.md`.

Neither prediction is supported. None of the eight market-quality coefficients has an interval excluding zero in the predicted direction, and three of the four activity coefficients are negative. This need not mean market quality is irrelevant, for three reasons. First, activity moves with time: in Los Angeles and Miami, trading grew steadily over the sample, so activity correlates with time at $r = 0.84$ and 0.66 and leaves little independent variation. Second, activity

responds to the forecast problem: trading and spreads react to the weather and to visible Kalshi–NBM disagreement, so even a detected association would not show that liquidity improves accuracy. Third, accurate prices may not require heavy trading. If a small number of attentive traders or market makers set prices at the margin, an accurate price can coexist with modest volume; this marginal-trader idea comes from studies of the Iowa Electronic Markets (Forsythe et al., 1992). Daily totals cannot settle the question, but trade-level evidence on how prices respond to individual trades (Hasbrouck, 1991) and to public forecast releases can, and the 5.5 million saved public trades make those tests possible.

The uncertainty control was not a hypothesis, but it is the clearest coefficient in the table. In Los Angeles and Miami, Kalshi's advantage is significantly *smaller* when NBM's distribution is wider: $-0.024$ and $-0.008$ per standard deviation. This is consistent with the location-bias account, although it does not establish it. A narrow NBM forecast centered in the wrong place is confidently wrong and scores badly, whereas a wide one hedges its misplacement, so the market's advantage is largest when the benchmark is sharp and off-center. This is an exploratory observation about a control variable.

**Step 6: Is It Just Timing? (P4 and S1)**

Kalshi's cutoff price might simply reflect information NBM had not yet absorbed. Two tests use the same 1,423 events and ask different questions. Both hold Kalshi fixed at its cutoff price and use the first NBM run archived after the cutoff, a median of 2.3–2.5 hours later (5.2 hours in Miami). P4 compares NBM with itself, the cutoff run against the later run, and asks whether NBM moves toward Kalshi and whether moving makes NBM more accurate; it speaks to whether the market anticipated where the public forecast would go. S1 compares the later NBM run directly with Kalshi's cutoff price and asks whether, given extra time, NBM catches up; it speaks to whether Kalshi's edge is only a few hours' head start.

**P4. First NBM revision.** I predicted that NBM would move toward the earlier Kalshi forecast and that the move would improve NBM's accuracy. Three measures keep these claims

separate: directional alignment (does the revision point toward Kalshi?), gap closure (does NBM end up closer to Kalshi?), and Brier gain (is the later NBM more accurate?). A revision can point toward Kalshi and still overshoot, so alignment does not imply closure. Every city's revisions move toward Kalshi, but none reliably ends closer to it, and accuracy improves clearly only in Phoenix (Table 9); normalized RPS improves in Los Angeles and Phoenix (Table A10). The prediction is partly supported. The aligned part of each revision is small, roughly 2–12% of the initial disagreement, so most of each revision is unrelated to Kalshi. Three explanations are possible, and these tests cannot separate them: NBM's revisions may mostly reflect new weather information unrelated to the market; a persistent station bias may survive from one run to the next; or "toward Kalshi" may partly mean "toward the truth," since Kalshi is closer to the outcome and any improving revision will appear to move toward it. NBM does not observe the market, so alignment does not mean NBM learns from Kalshi.

**S1. Fresh NBM versus Kalshi.** I predicted that Kalshi would keep its advantage. Against the later NBM run, Kalshi's cutoff price still has lower Brier loss in every city: 0.0214 in New York, 0.0629 in Los Angeles, 0.0118 in Miami, and 0.0074 in Phoenix, each with an interval above zero (Table 9). In Phoenix's smaller 124-event evaluation subset the interval includes zero (Table A11). Two to five more hours of public forecast information do not appear to erase the market's lead outside Phoenix, which argues against the simplest explanation, that Kalshi only sees information NBM has not yet released. Because the P4 means had been seen before this specification was fixed, S1 is exploratory.

**Table 9**
*First Post-Cutoff NBM Revision (P4) and Fresh-NBM Comparison (S1)*

| | New York | Los Angeles | Miami | Phoenix |
|---|---|---|---|---|
| Events | 198 | 539 | 501 | 185 |
| Median revision delay (h) | 2.5 | 2.3 | 5.2 | 2.3 |
| *P4: NBM revision* | | | | |
| Alignment | 0.0015 | 0.0013 | 0.0010 | 0.0013 |
| | [0.0007, 0.0022] | [0.0007, 0.0019] | [0.0006, 0.0013] | [0.0008, 0.0018] |
| Gap closure | −0.0004 | 0.0004 | −0.0004 | 0.0003 |
| | [−0.0020, 0.0009] | [−0.0007, 0.0015] | [−0.0010, 0.0003] | [−0.0006, 0.0013] |
| NBM Brier gain | 0.0005 | 0.0006 | −0.0003 | 0.0039 |
| | [−0.0028, 0.0037] | [−0.0012, 0.0025] | [−0.0024, 0.0018] | [0.0009, 0.0069] |
| *S1: later NBM minus cutoff Kalshi, Brier* | | | | |
| Gap | 0.0214 | 0.0629 | 0.0118 | 0.0074 |
| | [0.0128, 0.0307] | [0.0527, 0.0734] | [0.0075, 0.0163] | [0.0010, 0.0138] |

*Note.* Kalshi is held fixed at its cutoff price. The revision is the first NBM run archived after the cutoff and before the target window. Alignment is the mean product of the NBM revision and the initial Kalshi–NBM disagreement; gap closure is the reduction in mean squared distance to Kalshi; NBM Brier gain is cutoff minus later NBM loss (definitions in Appendix A). Positive values favor movement toward Kalshi, convergence, NBM improvement, and Kalshi, respectively. Individual 95% seven-event block-bootstrap intervals, 10,000 draws; no pooled or simultaneous city claim. Exploratory. RPS results and evaluation-subset sensitivities are in Tables A10 and A11.

### Step 7: Does the Advantage Survive a Location Correction? (P3 and Corrected P4/S1)

The location bias in Figure 1 is measured with outcomes a forecaster would not yet have seen. P3 asks whether a correction available at the time narrows the gap. At each cutoff it takes the median of (official high − NBM median) over earlier forecast–outcome pairs at the station whose official high had been published, conservatively timed at the report's issue clock plus one hour, with at least 60 such pairs. It shifts every NBM percentile by that amount, which keeps NBM's spread, rebuilds the six bins, and scores the result against the unchanged Kalshi price. Training can use NBM forecasts from before the market sample because no prices are needed, so every one of the 1,435 cutoffs is scored. The rule was fixed before any corrected score was computed, but the samples had already been inspected, so P3 is exploratory. I predicted that some of Kalshi's advantage would survive the correction, possibly in low-probability bins.

**Table 10**
*Past-Trained Location Correction (P3) and the Corrected Revision Repeat*

| | New York | Los Angeles | Miami | Phoenix |
|---|---|---|---|---|
| *P3 at the cutoff, full samples* | | | | |
| Events | 200 | 543 | 505 | 187 |
| Mean learned shift (°F) | +0.25 | +2.11 | +0.85 | −0.07 |
| Brier, Kalshi | 0.1024 | 0.1033 | 0.0978 | 0.0962 |
| Brier, raw NBM | 0.1247 | 0.1670 | 0.1093 | 0.1073 |
| Brier, corrected NBM | 0.1259 | 0.1312 | 0.1023 | 0.1077 |
| Correction gain | −0.0012 | 0.0358 | 0.0070 | −0.0004 |
| | [−0.0032, 0.0012] | [0.0251, 0.0464] | [0.0026, 0.0113] | [−0.0020, 0.0009] |
| Corrected NBM − Kalshi | 0.0235 | 0.0279 | 0.0044 | 0.0116 |
| | [0.0151, 0.0323] | [0.0206, 0.0354] | [0.0005, 0.0083] | [0.0051, 0.0179] |
| Same, evaluation subset | 0.0193 | 0.0298 | 0.0028 | 0.0120 |
| | [0.0097, 0.0293] | [0.0220, 0.0383] | [−0.0013, 0.0068] | [0.0040, 0.0192] |
| *Corrected revision repeat (same shift on both NBM runs)* | | | | |
| Pairs | 198 | 539 | 501 | 185 |
| Alignment toward Kalshi | 0.0015 | 0.0016 | 0.0011 | 0.0013 |
| | [0.0007, 0.0022] | [0.0012, 0.0021] | [0.0008, 0.0014] | [0.0009, 0.0019] |
| Gap closure | −0.0004 | 0.0010 | 0.0001 | 0.0004 |
| | [−0.0019, 0.0010] | [0.0001, 0.0018] | [−0.0004, 0.0006] | [−0.0004, 0.0014] |
| NBM revision Brier gain | 0.0004 | 0.0016 | 0.0020 | 0.0046 |
| | [−0.0027, 0.0035] | [−0.0003, 0.0034] | [0.0005, 0.0036] | [0.0015, 0.0079] |
| Later corrected NBM − Kalshi | 0.0226 | 0.0263 | 0.0025 | 0.0069 |
| | [0.0138, 0.0316] | [0.0190, 0.0338] | [−0.0015, 0.0065] | [0.0007, 0.0129] |

*Note.* Primary expanding-window rule. The learned shift is official high minus NBM median over earlier published outcomes; it changes from date to date, and its mean is shown. Positive gains mean the correction or revision made NBM more accurate; positive gaps favor Kalshi. Evaluation subsets are New York's 139 held-out dates and the three cities' combination-evaluation dates (482, 444, 126). Individual 95% seven-event block-bootstrap intervals, 10,000 draws, conditional on the learned shifts; no pooled or simultaneous city claim. Results: `output/weather/p3_station_correction_2026-10-05/` and `output/weather/corrected_nbm_revision_2026-10-05/`.

**P3. Location correction at the cutoff.** The correction makes NBM more accurate in Los Angeles and Miami and narrows the Kalshi gap there from 0.0637 to 0.0279 and from 0.0115 to 0.0044, roughly 56% and 61% (Table 10). It does not clearly help in New York or Phoenix, and on New York's 139 held-out dates it makes NBM slightly worse (−0.0023 [−0.0038, −0.0010]). In every full sample Kalshi still has lower loss than corrected NBM; in Miami's smaller evaluation subset the interval includes zero, which is not evidence that the two are equal. A last-60-pair rolling window gives the same conclusions, and timing buffers of zero or two hours

on report availability change no shift. Normalized RPS favors Kalshi in New York, Los Angeles, and Phoenix, but its Miami intervals include zero. The prediction that some advantage survives is supported in the full samples. Whether what survives sits in low-probability bins is the separate test P6.

Two cautions keep this from being read as a measured "station-bias share." First, the learned shift is whatever error persisted in earlier outcomes, which can mix the grid-versus-station difference with season, the NBM version change, and the gap between NBM's 18-hour window and the settlement day; P3 asks whether a feasible correction closes the gap, not which physical cause produced the error. Second, the New York result shows that the in-sample bias in Figure 1 was not available in advance: the shift learned from earlier New York outcomes averaged +0.25°F, the opposite sign of the error over the held-out months, and it shrank toward zero over the sample. What the market prices in New York, and the remaining advantage everywhere, is therefore not explained by a simple past-trained location correction.

**Corrected P4 and S1.** Repeating the revision tests with the same cutoff shift applied to both NBM runs leaves the earlier pattern mostly intact (Table 10, lower panel). Revisions still point toward the earlier Kalshi forecast in every city; they now end measurably closer to it only in Los Angeles, and that closure is inconclusive in its smaller subset. The revision improves corrected NBM in Miami and Phoenix. Against the later corrected run, Kalshi's cutoff price keeps a lower loss in New York, Los Angeles, and Phoenix; in Miami the interval includes zero, as it does in the smaller Miami and Phoenix subsets. In Miami, then, a corrected NBM run issued about five hours after the cutoff is statistically indistinguishable from the market. The later run has information the cutoff price could not, so this does not overturn the cutoff comparison. Because the correction is applied identically to both runs, these measures cannot show that NBM learns from the market.

**Step 8: Does Kalshi Add to Six Public Forecasts? (P1/P2)**

The public benchmark now combines six sources available by the original cutoff. Dense NBM contributes its median, GEFS and direct-public ECMWF contribute the mean of member-specific maxima, NDFD and the local NWS Point Forecast Matrix (PFM) contribute native daytime highs, and HRRR contributes its maximum sampled hourly temperature over 12 UTC through 05 UTC the following day. These products cover different physical periods and locations. The model learns their relationship to the settlement station's high from earlier events.

For each city, a Gaussian predictive distribution has mean $\mu_i = \alpha + \sum_{j=1}^{6} w_j f_{ij}$ and variance $\sigma_i^2 = a + bS_i^2$. Source weights are nonnegative and sum to one. The spread predictor averages GEFS and ECMWF member variances with the NBM normal-scale variance inferred from its 10th and 90th percentiles. Parameters minimize continuous ranked probability score on the latest 60 eligible events whose completed-day station reports were available before the current cutoff. Report availability uses the later body/WMO issue clock plus one hour. Each fit uses the latest report version then available. Integrating the fitted curve over the original contract boundaries gives six probabilities, including both open tails.

The second forecast combines these probabilities with Kalshi, using $(1 - \lambda)p_{\text{public}} + \lambda p_{\text{Kalshi}}$. The market weight lies between zero and one and minimizes past Brier loss. To avoid rewarding public forecasts fitted to their own training outcomes, the 60 training events are split into three consecutive groups of 20. Each group's public probabilities come from a model fitted to the other 40. These internal predictions use only information available at the current outer cutoff, although a training fold may contain dates later than an internally held-out row. The public model is refitted to all 60 before predicting the next event. The evaluated event's outcome enters neither fit.

The specification was frozen on November 4 before these estimates, following the user's execution request and earlier prediction that public aggregation would narrow, but incompletely eliminate, Kalshi's advantage. Earlier NBM findings had already informed this follow-up, which

is exploratory. Lang et al. (2020) support adaptive training as a methodological motivation. The 60-event window was chosen before this run, with 40-event and expanding-window checks.

All three variants evaluated the same 1,184 events, with zero fitting failures. Of 1,435 original events, 247 lacked sufficient earlier eligible outcomes and four had an ECMWF availability proxy after cutoff. New York contributes 138 events, Los Angeles 481, Miami 439, and Phoenix 126. New York retains its fixed 02 UTC cutoff; the other cities retain 22 local time on the preceding date. Probabilities are scored against the original market settlement indicators. Training uses station reports available at the time, allowing discrepancies between historical report versions and final settlement.

The primary audit checked all 71,040 training memberships and found no label discrepancies with final settlement in those memberships. The variance intercept reached a constraint boundary in 638 of 1,184 public fits, and at least one source weight reached a boundary in 1,158 fits. The constrained model often excludes a source. Boundary frequency is a model diagnostic and limits interpretation of individual source weights. The original evaluation subsets and both alternative windows are reported in Appendix C.

**Table P1**
*Mean Brier Loss on Identical Evaluation Events*

| City | Events | NBM | Public | Kalshi | Blend |
|---|---|---|---|---|---|
| New York | 138 | 0.1168 | 0.1099 | 0.1002 | 0.0999 |
| Los Angeles | 481 | 0.1687 | 0.1097 | 0.1018 | 0.1019 |
| Miami | 439 | 0.1086 | 0.1001 | 0.0995 | 0.0982 |
| Phoenix | 126 | 0.1055 | 0.1009 | 0.0944 | 0.0953 |

*Note.* Lower loss is better. Public denotes the six-source Gaussian model. Blend denotes public plus Kalshi with a weight trained on earlier events. Each event contributes one mean six-bin Brier loss.

**Table P2**
*Paired Brier Differences and 95% Intervals*

| City | Public minus NBM | Public minus Kalshi | Public minus blend |
|---|---|---|---|
| New York | -0.0069 [-0.0132, -0.0004] | 0.0097 [0.0012, 0.0181] | 0.0101 [0.0032, 0.0169] |
| Los Angeles | -0.0590 [-0.0695, -0.0491] | 0.0079 [0.0034, 0.0126] | 0.0078 [0.0042, 0.0117] |
| Miami | -0.0085 [-0.0135, -0.0034] | 0.0006 [-0.0034, 0.0045] | 0.0019 [-0.0008, 0.0047] |
| Phoenix | -0.0045 [-0.0141, 0.0047] | 0.0065 [-0.0010, 0.0149] | 0.0057 [-0.0003, 0.0121] |

*Note.* Negative values favor public over the named comparator; positive public-minus-blend values favor adding Kalshi. Intervals use 10,000 paired circular bootstrap draws with seven consecutive event-days per block, separately by city. They condition on saved fitted predictions. Intervals are pointwise across the reported comparisons.

The combined public forecast improves on original NBM in New York, Los Angeles and Miami under the primary Brier intervals. Phoenix's interval includes zero. Kalshi alone retains lower loss than public in New York and Los Angeles, with intervals excluding zero. Miami and Phoenix have positive point estimates for Kalshi's advantage but intervals spanning zero. Adding Kalshi to public improves the primary forecast in New York and Los Angeles. Its smaller estimated gains in Miami and Phoenix are inconclusive by the primary Brier intervals. The blend's average market weights are 0.736, 0.711, 0.642 and 0.829 in New York, Los Angeles, Miami and Phoenix, respectively. A positive trained weight does not by itself establish an evaluation gain. These comparisons also leave open whether public information improves on Kalshi alone.

The secondary ranked probability score gives the same main New York and Los Angeles findings. Phoenix's public-minus-blend RPS interval is narrowly positive, unlike its Brier interval. The 40-event check makes New York's Kalshi and blending gains inconclusive, while supporting Phoenix's gains and a small Miami blending gain. The expanding check retains the New York and Los Angeles findings, but its New York public model no longer improves on original NBM. Los Angeles provides the most consistent evidence across training-window choices. These changes limit claims that every city exhibits the same mechanism.

A separately calibrated PFM forecast improves on raw NBM clearly only in Los Angeles. It performs worse than P3-corrected NBM in Miami, while the other PFM-versus-P3 Brier

intervals include zero. PFM's calibration uses the same past window, an estimated temperature correction and a constant Gaussian variance. These results only partly support the earlier PFM prediction.

Forecast gains establish predictive content relative to this specified public model. Public products use recorded archive or issuance proxies, and several predict a different physical period from the settlement day. Those limits, shared weather inputs, and the restricted Gaussian model leave room for alternative public explanations. The bulletin and liquidity studies remain separate mechanisms to test. The estimates do not identify private signals or trader types.

**Capstone Mechanism: Public Information and Market Liquidity**

The public-forecast results motivate two competing economic forces. Less informative public guidance gives markets more room to improve forecasts, while deeper and cheaper trading may increase the incentive and opportunity to incorporate information. City size can move both forces simultaneously. The November 5 prediction is that liquidity helps most when public forecasts are less informative. This prediction remains to be estimated under a complete specification.

The proposed historical outcome is the public-minus-Kalshi Brier loss, with the public-minus-blend gain as a separate comparison. The design uses observed historical cutoff spreads within each city, interacting liquidity with a measure of public weakness available before the cutoff. Earlier forecast losses are a candidate measure of weakness. Public uncertainty measures a distinct feature, and current realized public loss would create a mechanical relationship because it is already a component of the dependent variable. Actual budgets and local staffing have not been measured. The four cities cannot identify a broad causal effect of local forecasting resources.

The live collector supplies prospective evidence on what spread measures capture. The September 21 project log records its initial daily implementation. A November 5 audit replayed first, middle and last snapshots from 74 saved event folders, covering 222 snapshots and 1,332

contract books. All six contracts were present in every sampled snapshot and every replayed numerical metric matched the saved summary. Empty YES-bid ladders occurred in 284 sampled books. Thus six active contracts do not guarantee six two-sided midpoints. The earlier count of six qualifying city-events described a restrictive complete-midpoint sample;56 of 66 elapsed event windows in the November 4 inventory had sufficient near-cutoff polling. Fifty failed quote-side completeness despite sufficient polling. General liquidity analysis should preserve one-sided books and report metric-specific coverage.

The live period and historical P1/P2 event sample do not overlap. Interpreting historical spreads using live depth therefore requires stability conditional on city, price and forecast horizon. This validation remains distinct from a historical depth reconstruction. The resource hypothesis motivates the analysis; direct tests of budgets or staffing require additional measurements. The accepted bulletin specification provides another information-channel test, measuring movement from one minute before to five minutes after generation, with separate-city estimates. Its execution remains on hold for validated NYC historical minute candles.

**Planned Tests**

The completed tests now include a six-source public benchmark and an incrementally trained Kalshi blend. Their city-specific results and sensitivities appear in Step 8. The planned tests ask whether anything remains once stronger public explanations are removed. Each will be specified, including its unit, cutoff, information set, predicted sign, sample rule, dependence structure, and interpretation, before its outcome is examined. Table 11 lists them; the most developed are described below and in Appendix B.

**Table 11**
*Planned Tests*

| Test | Question | Data status | Would not establish |
|---|---|---|---|
| P3b Location and spread | Does also recalibrating NBM's spread on past outcomes close more of the gap? | P3 shifts saved; scale rule not yet specified | That remaining gains are informational |
| S5 Bias split | How much of Kalshi's shift from NBM does the past-trained bias predict? | P3 shifts saved | Private information |
| Revision decomposition | Do NBM revisions move toward Kalshi beyond moving toward the outcome? | Revision pairs validated | That NBM responds to Kalshi |
| S3 Brier decomposition | Is Kalshi's edge in resolution or reliability, and does recalibration change it? | Data on hand; draft specification | Trader identity |
| P6 Low-probability correction | Among bins a corrected NBM rates ≤5%, do market prices separate winners? | Corrected NBM constructed | Trader identity |
| S6 Neighbor-station placebo | Does the edge vanish at a nearby non-settlement station? | Sampled neighbor highs built; outcome definition open | Private data use |
| S7 Evening observations | Did market moves follow public station observations? | ASOS saved | Private data use |
| S10 Release response | Do prices move after public forecast releases? | Rules accepted; three-city panel prepared; awaiting NYC minute source | Causal trader response |
| S9 Profitability | Could a public strategy profit after fees? | Fee schedules saved, not yet read | Welfare |

**What does the correction leave (P3b, S5)?** P3 shifts NBM's location but keeps its spread. P3b would also recalibrate the spread from past outcomes, in the spirit of Brown (2026), which also tests whether the exploratory Los Angeles and Miami pattern in Table 6, a larger gap when NBM is narrow, reflects an overconfident benchmark. S5 then splits Kalshi's shift away from NBM into the part predicted by the saved past-trained shift and a residual. Neither would identify a causal station-bias share.

**Do revisions anticipate the market or the outcome?** P4 found revisions that move toward Kalshi without converging. Because Kalshi is closer to the outcome, any improving revision will look like movement toward it. The planned decomposition relates each NBM revision both to the initial Kalshi–NBM disagreement and to the initial NBM error, asking whether a Kalshi component survives once the outcome direction is accounted for. It can be run on both raw and station-corrected NBM; the corrected repeat in Table 10 showed that alignment survives the correction, which the decomposition would then explain.

**Stronger public forecasts (P1, P2), completed.** Step 8 reports the six-source model and the simple convex probability blend chosen for this capstone. A separate unrestricted encompassing design (S8) remains open.

**Does the market selectively correct unlikely outcomes (P6)?** The working definition selects bins assigned at most 5% by a past-trained, location-corrected NBM distribution (10% as sensitivity), fixing membership before any later movement or outcome. Correcting first matters; otherwise the test would rediscover the location bias in the tails. T4's longshot result makes this demanding, because uniform upward revision would worsen forecasts, so both increases and decreases must be evaluated, including cases where the selected bin loses (Lerch et al., 2017). Earlier outcomes have already informed this hypothesis, so a test on the same data will be labeled exploratory.

**What can the data identify (S6, S7, S10)?** A neighbor-station placebo, evening station observations, and price responses to public forecast releases could show whether market moves follow public data. Even so, historical quotes do not identify the ratio of informed to speculative traders or reveal private signals. The feasible contribution is to locate where the market's predictive gains arise and to test competing public explanations.

## Conclusion

The operational question has a qualified answer. At a 10 p.m. cutoff, Kalshi's six-bin probabilities have lower loss than the reconstructed NBM distribution in all four cities. In 139

held-out New York events the reduction is 14.5%, and a chronologically trained combination places all weight on Kalshi; in the three other cities the comparisons are exploratory but point the same way, and mixing in NBM does not improve on Kalshi. The advantage persists against an NBM run released two to five hours after the cutoff, except in Phoenix's smaller subset, and NBM's later runs move toward the market without converging to it. Sparse percentile reconstruction, target-window mismatch, and market activity do not appear to explain the result, and it coexists with overpriced longshots.

The descriptive evidence suggests where part of the advantage comes from. Kalshi's forecast is better centered rather than sharper. Over the sample, NBM's median at each settlement station was off-center, by about 1–2°F in Los Angeles and New York, and Kalshi's distribution sat on the correcting side. Kalshi's accuracy is stable across cities while NBM's is not, and in Los Angeles and Miami the advantage is largest on days when NBM is confidently off-center. A correction learned only from earlier outcomes confirms part of this account: it removes more than half of the gap in Los Angeles and Miami. It also limits it: the correction does not help in New York, where the earlier error pointed the other way, or in Phoenix, and Kalshi retains lower loss than the corrected benchmark in every full sample. The market may be pricing station-level errors faster or more flexibly than a simple past-trained shift, or drawing on other public information; these data cannot yet tell which.

The six-source extension narrows the original NBM gap. In the primary 60-event model, Kalshi retains an advantage and adds predictive value in New York and Los Angeles, while Miami and Phoenix are inconclusive by Brier intervals. Training-window checks qualify the New York result and make Los Angeles the most consistent case. Stronger public guidance therefore changes the breadth of the claim. The remaining release-response and liquidity analyses will examine when the residual predictive content appears.

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# Appendix A

## Methods, Measurement Details, and Supporting Tables

### Construction and Scoring

For each bin, the unnormalized midpoint is $(\text{bid} + \text{ask})/2$; division by the sum of the six midpoints produces the market distribution. Quotes satisfy the cutoff and age rule individually and need not be simultaneous, so normalization can transmit one bin's quote movement to the others. NBM quantiles are converted to Fahrenheit, repaired by equal-weight isotonic regression when crossed, and inverted using linear interpolation with a bounded linear tail rule; half-degree thresholds map the continuous approximation to integer-temperature bins. The normalized ranked probability score is

$$\text{RPS}_i(p) = \frac{1}{5}\sum_{k=1}^{5}\left(\sum_{j=1}^{k} p_{ij} - \sum_{j=1}^{k} y_{ij}\right)^2. \tag{3}$$

Dividing Brier loss by six and RPS by five fixes the reporting scale; neither scaling changes which forecast performs better.

### Inference

One resolved event is one observation; its six bins, and any repeated forecasts of it, stay together. For each city and sample, the mean paired loss difference is resampled with a circular moving-block bootstrap: blocks of seven consecutive eligible events, wrapping at the end of the sample, 10,000 replications, with a fixed seed. Intervals are 95% percentile intervals with linear interpolation. Two-sided *p*-values come from the same blocks applied to differences centered at zero, with a plus-one correction. Seven eligible events may span more than seven calendar days where dates are missing. A Bartlett-kernel HAC standard error with seven lags provides a secondary check for Brier differences. Cities are estimated separately and intervals are individual and unadjusted; the paper makes no pooled or simultaneous four-city claim. Within a city, all measures for a test share the same resampled events.

**Forecast Combination**

For each target date, every candidate weight $\lambda \in \{0, 0.05, \ldots, 1\}$ is scored on the same earlier events whose official outcomes were available before that target's cutoff, and the weight with the lowest mean Brier loss is applied to the target. Ties favor the smaller Kalshi weight. The window expands as outcomes settle, with a minimum of 60 training events, and each target is scored without its own outcome. The three-city version applies these rules unchanged.

**Revision Measures**

For event $i$, let $K_i$ be the fixed cutoff Kalshi vector, $N_{0i}$ the cutoff NBM vector, and $N_{1i}$ the first NBM vector archived after the cutoff and before the target window. Averaging over the six bins,

$$\begin{aligned} \text{Alignment:} \quad & A_i = \overline{(N_{1i} - N_{0i})(K_i - N_{0i})}, \\ \text{Gap closure:} \quad & G_i = \overline{(N_{0i} - K_i)^2} - \overline{(N_{1i} - K_i)^2} = 2A_i - \overline{(N_{1i} - N_{0i})^2}, \\ \text{NBM Brier gain:} \quad & C_i = B_i(N_0) - B_i(N_1). \end{aligned}$$

Positive alignment means the revision has a component toward the earlier disagreement; positive closure means NBM ends closer to Kalshi; positive gain means the later NBM is more accurate. The identity for $G_i$ shows why a revision can align ($A_i > 0$) yet fail to close the gap if it is large relative to its aligned part. Ties in the first eligible revision are broken by the earlier model cycle, without reference to scores. S1 uses the same events and defines the gap as $B_i(N_1) - B_i(K)$.

**Location Bias Measurement**

Location bias is the official daily high minus the NBM 50th percentile at the station's nearest grid point, at the cutoff vintage. These are full-sample descriptive averages, not corrections: they use outcomes that a real-time forecaster would not yet have seen. Intervals are 2,000-draw seven-event block-bootstrap intervals used for description only. Table A9 compares the exact-high measure with the settled-bin-midpoint proxy used in an earlier draft.

## Validation

Each dataset was checked for unique keys, valid time ordering, plausible values, expected coverage, exhaustive bins, and probabilities summing to one before any score was computed. Raw files were preserved and each construction was written to a new dated output. For the revision tests, an independent script verified event, bin, outcome, cutoff, and station identities; the physical target window; first-revision selection; and the percentile grids. It rebuilt all 2,858 NBM curves with separate code (maximum bin difference $8.4 \times 10^{-14}$) and rehashed all 98,984 raw files (163 GB) against their recorded checksums and download records. Each test's specification was fixed after validation and before estimation, and an independent script rechecked every reported mean and interval afterward.

**Table A1**
*Market Inputs by Sample*

| | New York | Los Angeles | Miami | Phoenix |
|---|---|---|---|---|
| Mean bin spread, mean / max (cents) | 1.42 / 3.17 | 2.48 / 41.83 | 2.75 / 14.50 | 1.82 / 6.33 |
| Raw midpoint sum, mean | 1.034 | 1.055 | 1.050 | 1.039 |
| Raw midpoint sum, range | 0.96–1.11 | 0.91–1.98 | 0.94–1.26 | 0.88–1.14 |
| Oldest selected quote, median (min) | 0 | 1 | 4 | 14 |
| Contracts traded to cutoff, median | 17,108 | 16,620 | 8,098 | 7,419 |

*Note.* New York volume sums hourly candles from market open; city volume sums observed one-minute candles to the cutoff, so levels are only roughly comparable.

**Table A2**
*Mean Training Brier Loss by Kalshi Weight*

| Target date (training events) | $\lambda = 0$ | 0.5 | 0.8 | 0.9 | 0.95 | 1 |
|---|---|---|---|---|---|---|
| April 11 (60) | 0.1429 | 0.1171 | 0.1096 | 0.1084 | 0.1081 | 0.1079 |
| August 27 (198) | 0.1251 | 0.1085 | 0.1037 | 0.1029 | 0.1027 | 0.1026 |

**Table A3**
*New York City Calibration by Probability Group*

| | Kalshi | | | NBM | | |
|---|---|---|---|---|---|---|
| Stated probability | Bins | Mean (%) | Realized (%) | Bins | Mean (%) | Realized (%) |
| 0–5% | 345 | 1.8 | 0.3 | 275 | 0.8 | 0.7 |
| >5–10% | 89 | 6.9 | 4.5 | 108 | 7.6 | 8.3 |
| >10–20% | 101 | 15.2 | 15.8 | 141 | 15.3 | 19.9 |
| >20–40% | 192 | 30.2 | 31.3 | 231 | 28.4 | 26.4 |
| >40–60% | 92 | 46.0 | 48.9 | 63 | 47.1 | 46.0 |
| >60–100% | 15 | 73.8 | 86.7 | 16 | 74.2 | 62.5 |

*Note.* ECE 1.75 points (Kalshi) and 1.74 (NBM); root-mean-square calibration error 2.40 and 2.72.

**Table A4**
*New York City Held-Out Brier Scores by Month*

| Month (2026) | Days | Mean high (°F) | Kalshi | NBM | Difference | Kalshi wins |
|---|---|---|---|---|---|---|
| April (from 11th) | 20 | 66.4 | 0.1071 | 0.1334 | 0.0263 | 70.0% |
| May | 31 | 71.5 | 0.0948 | 0.1292 | 0.0344 | 67.7% |
| June | 30 | 81.4 | 0.1085 | 0.1223 | 0.0138 | 60.0% |
| July | 31 | 83.7 | 0.0947 | 0.0970 | 0.0023 | 48.4% |
| August (to 27th) | 27 | 83.1 | 0.0979 | 0.1083 | 0.0104 | 55.6% |

**Table A5**
*New York City Observed-Window Agreement Audit*

| Comparison | Agree | Dates | Share | 95% CI |
|---|---|---|---|---|
| Same maximum, contract vs. NBM window | 186 | 200 | 0.930 | [0.886, 0.958] |
| Maxima within 1°F | 187 | 200 | 0.935 | [0.892, 0.962] |
| Same contract bin | 194 | 200 | 0.970 | [0.936, 0.986] |
| Recorded settlement = observed contract-window max | 112 | 199 | 0.563 | [0.493, 0.630] |
| Recorded settlement within 1°F of observed max | 192 | 199 | 0.965 | [0.929, 0.983] |

*Note.* Contract window [05:00, 05:00) UTC; NBM window [12:00, 06:00) UTC; dates require ≥90% hourly coverage in both. Observations are sampled station reports, which can miss the official peak, so the last two rows describe the observation series, not settlement errors.

**Table A6**
*Diagnostics for the Original Joint Market-Quality Regressions (Table A8)*

| | Brier model | RPS model |
|---|---|---|
| $R^2$ (adjusted) | 0.039 (0.011) | 0.071 (0.043) |
| Dates with high leverage / high Cook's distance | 7 / 10 | 7 / 7 |
| Largest absolute studentized residual | 3.36 | 6.29 |
| Breusch–Pagan / RESET / Ljung–Box (7) *p* | 0.40 / 0.82 / 0.79 | 0.03 / 0.19 / 0.14 |
| Shapiro–Wilk *p* | 0.58 | < 0.001 |

*Note.* Screening tests use classical reference distributions; block-bootstrap intervals in Table A8 remain primary. No dates were removed.

**Table A7**
*City Archive-Timing Sensitivity*

| | No buffer | | 15-minute buffer | | 60-minute buffer | |
|---|---|---|---|---|---|---|
| City | *n* | Brier gap | *n* | Brier gap | *n* | Brier gap |
| Los Angeles | 543 | 0.0628 | 543 | 0.0628 | 539 | 0.0627 |
| Miami | 505 | 0.0116 | 470 | 0.0132 | 228 | 0.0126 |
| Phoenix | 187 | 0.0113 | 187 | 0.0113 | 183 | 0.0109 |

*Note.* A buffer drops events whose NBM file was archived less than that many minutes before the cutoff. Post-score provenance diagnostic; the archive time does not verify first publication.

**Table A8**
*Original Joint Market-Quality Model, New York (September 21 Specification)*

| | Brier difference | | RPS difference | |
|---|---|---|---|---|
| Predictor (per SD) | Estimate | 95% CI | Estimate | 95% CI |
| Log trading volume | −0.0022 | [−0.0118, 0.0072] | −0.0065 | [−0.0252, 0.0085] |
| Mean cutoff spread | 0.0007 | [−0.0093, 0.0081] | −0.0018 | [−0.0184, 0.0102] |
| NBM P90–P10 width | 0.0000 | [−0.0089, 0.0091] | 0.0033 | [−0.0090, 0.0171] |
| Date trend | −0.0110 | [−0.0214, 0.0010] | −0.0239 | [−0.0395, −0.0074] |
| $R^2$ | | 0.039 | | 0.071 |
| *Robustness: log trade count in place of volume* | | | | |
| Log trade count | −0.0028 | [−0.0128, 0.0058] | −0.0082 | [−0.0274, 0.0048] |
| *Unadjusted top-minus-bottom quartile contrasts* | | | | |
| Volume Q4 − Q1 | 0.0141 | [−0.0023, 0.0316] | 0.0313 | [0.0035, 0.0619] |
| Spread Q4 − Q1 | −0.0121 | [−0.0365, 0.0097] | −0.0282 | [−0.0699, 0.0090] |
| NBM width Q4 − Q1 | 0.0062 | [−0.0151, 0.0309] | 0.0246 | [−0.0136, 0.0678] |

*Note.* $n = 139$. Outcome is NBM loss minus Kalshi loss. All four predictors enter one model, so this is not the same as Table 6's separate models; residual diagnostics are in Table A6. Intervals from 10,000 seven-day block-bootstrap draws. Quartile contrasts are descriptive, not adjusted for the other predictors or for multiple comparisons.

**Table A9**
*NBM Location Bias by Outcome Measure*

| City | Measure | Events | Mean (°F) | 95% CI |
|---|---|---|---|---|
| Los Angeles | Official high, all events | 543 | 1.77 | [1.28, 2.22] |
| | Climate-report high only | 541 | 1.77 | [1.29, 2.21] |
| | Settled-bin midpoint, interior bins | 460 | 1.92 | [1.59, 2.24] |
| Miami | Official high, all events | 505 | 0.70 | [0.54, 0.86] |
| | Climate-report high only | 499 | 0.71 | [0.55, 0.87] |
| | Settled-bin midpoint, interior bins | 465 | 0.58 | [0.43, 0.72] |
| Phoenix | Official high, all events | 187 | −0.30 | [−0.55, −0.04] |
| | Climate-report high only | 187 | −0.30 | [−0.55, −0.04] |
| | Settled-bin midpoint, interior bins | 180 | −0.30 | [−0.53, −0.05] |

*Note.* Outcome minus the NBM median; positive values mean NBM was too cold. "Official high, all events" fills eight missing climate-report highs with retrospective ACIS values, which were not available at the cutoff. The midpoint proxy omits open-bin outcomes; including them lowers the Los Angeles mean and raises the Miami mean. New York uses the reported high on 139 held-out dates (−1.01°F [−1.51, −0.48]).

**Table A10**
*First NBM Revision: Secondary Score and Sensitivity Samples*

| | New York | Los Angeles | Miami | Phoenix |
|---|---|---|---|---|
| *Full samples: NBM normalized-RPS gain* | | | | |
| Events | 198 | 539 | 501 | 185 |
| RPS gain | 0.0038 | 0.0035 | 0.0007 | 0.0039 |
| | [0.0001, 0.0080] | [0.0014, 0.0056] | [−0.0012, 0.0026] | [0.0006, 0.0073] |
| Centered-null *p* | 0.069 | 0.001 | 0.474 | 0.024 |
| *Evaluation subsets: NBM Brier gain* | | | | |
| Events | 137 | 478 | 440 | 124 |
| Alignment | 0.0015 | 0.0013 | 0.0010 | 0.0014 |
| Brier gain | 0.0011 | 0.0003 | −0.0008 | 0.0040 |
| | [−0.0026, 0.0048] | [−0.0017, 0.0022] | [−0.0030, 0.0015] | [0.0003, 0.0079] |

*Note.* Evaluation subsets are New York's original held-out dates and the three cities' combination-evaluation dates that have an eligible revision. All alignment intervals in the subsets exclude zero; all gap-closure intervals include zero. Percentile intervals and centered-null tests are not exact inversions, which is why New York's RPS interval excludes zero while its *p*-value exceeds 0.05.

**Table A11**

*Fresh-NBM Comparison: Secondary Score and Sensitivity Samples*

| | New York | Los Angeles | Miami | Phoenix |
|---|---|---|---|---|
| *Full samples: later NBM minus Kalshi, normalized RPS* | | | | |
| Gap | 0.0453 | 0.1072 | 0.0139 | 0.0061 |
| | [0.0301, 0.0616] | [0.0902, 0.1250] | [0.0083, 0.0199] | [−0.0002, 0.0122] |
| *Evaluation subsets: later NBM minus Kalshi, Brier* | | | | |
| Events | 137 | 478 | 440 | 124 |
| Gap | 0.0154 | 0.0669 | 0.0111 | 0.0074 |
| | [0.0065, 0.0248] | [0.0564, 0.0779] | [0.0065, 0.0159] | [−0.0009, 0.0150] |
| Share of events Kalshi better (full) | 63% | 75% | 58% | 56% |

*Note.* Positive values favor Kalshi. HAC tests support each full-sample Brier difference. Phoenix's evaluation subset ($p = 0.066$) and full-sample RPS interval are inconclusive rather than evidence for NBM.

## Appendix B

## Specifications for the Planned Tests

These analyses are proposed. Their final rules will be fixed before outcomes are examined.

**Past-trained station correction (P3), as implemented.** Completed October 5, 2026 under the frozen specification `forecast_evaluation/NBM_STATION_CORRECTION_SPECIFICATION_2026-10-04.md`; results in Table 10. For target $i$ with cutoff $t_i$, let $\hat{b}_i$ be the median of (official high − NBM median) over earlier forecast–outcome pairs at that station whose official high was published before $t_i$; a mean-error version is a sensitivity. Training pairs need not have Kalshi markets, so the window can begin before the market sample. The primary window expands; a rolling window of the most recent 60 pairs is a sensitivity, and a minimum of 60 usable pairs is required. Report availability is the issue clock plus one hour, with zero- and two-hour sensitivities. Every NBM quantile is shifted by $\hat{b}_i$, preserving the spread, the six-bin distribution is rebuilt, and corrected and raw NBM are each scored against Kalshi on identical dates with the inference in Appendix A. Retrospective ACIS values are excluded from training because they were not public at the time. For the revision tests, the correction learned at the cutoff is applied to both NBM vintages.

**Revision decomposition.** For event $i$ and bin $j$, relate the revision $\Delta_{ij} = N_{1ij} - N_{0ij}$ jointly to the initial disagreement $K_{ij} - N_{0ij}$ and the initial error $y_{ij} - N_{0ij}$, with inference clustered by event and block-bootstrapped over time. Because both regressors share $N_{0ij}$, a noisy or biased benchmark will show partial alignment through mean reversion; the informative quantity is a Kalshi component beyond the outcome direction, estimated first on raw and then on station-corrected NBM.

**Public-only benchmarks (P1, P2), as implemented.** Completed November 4 under the frozen specification and reported in Step 8. Public weights lie in $[0, 1]$ and sum to one. The Kalshi probability weight also lies in $[0, 1]$. All variants use identical eligible dates. Separate unrestricted encompassing tests remain unestimated.

**Brier decomposition (S3).** Decompose each forecast's Brier score into reliability, resolution, and uncertainty with coarse, prespecified probability groups and block-bootstrapped intervals, to ask whether Kalshi's advantage lies in resolution, in reliability, or in both.

**Selective low-probability correction (P6).** Select bins with corrected initial NBM probability $\leq 0.05$ (sensitivity $\leq 0.10$) before observing later quotes or outcomes. Compare realized frequencies across groups defined by the market's level or change, and compare held-out loss on the selected set between public-only and public-plus-market forecasts. Report occurrence counts, distinguish summed bin losses from a score for the union of selected bins, and preserve within-event dependence.

# Appendix C

## P1/P2 Supporting Results

**Table C1**
*Rolling 60; full usable*

| City | Public minus NBM | Public minus Kalshi | Public minus blend |
|---|---|---|---|
| New York (138) | -0.0069 [-0.0132, -0.0004] | 0.0097 [0.0012, 0.0181] | 0.0101 [0.0032, 0.0169] |
| Los Angeles (481) | -0.0590 [-0.0695, -0.0491] | 0.0079 [0.0034, 0.0126] | 0.0078 [0.0042, 0.0117] |
| Miami (439) | -0.0085 [-0.0135, -0.0034] | 0.0006 [-0.0034, 0.0045] | 0.0019 [-0.0008, 0.0047] |
| Phoenix (126) | -0.0045 [-0.0141, 0.0047] | 0.0065 [-0.0010, 0.0149] | 0.0057 [-0.0003, 0.0121] |

*Note.* Brier loss differences with pointwise 95% paired seven-event bootstrap intervals. Counts appear in parentheses. Windows share the same full evaluation sample.

**Table C2**
*Rolling 60; original evaluation subset*

| City | Public minus NBM | Public minus Kalshi | Public minus blend |
|---|---|---|---|
| New York (138) | -0.0069 [-0.0133, -0.0004] | 0.0097 [0.0012, 0.0183] | 0.0101 [0.0033, 0.0169] |
| Los Angeles (481) | -0.0590 [-0.0697, -0.0489] | 0.0079 [0.0034, 0.0126] | 0.0078 [0.0042, 0.0116] |
| Miami (439) | -0.0085 [-0.0136, -0.0036] | 0.0006 [-0.0032, 0.0045] | 0.0019 [-0.0008, 0.0047] |
| Phoenix (126) | -0.0045 [-0.0141, 0.0048] | 0.0065 [-0.0010, 0.0147] | 0.0057 [-0.0004, 0.0121] |

*Note.* Brier loss differences with pointwise 95% paired seven-event bootstrap intervals. Counts appear in parentheses. Windows share the same full evaluation sample.

**Table C3**
*Rolling 40; full usable*

| City | Public minus NBM | Public minus Kalshi | Public minus blend |
|---|---|---|---|
| New York (138) | -0.0109 [-0.0186, -0.0030] | 0.0058 [-0.0023, 0.0139] | 0.0059 [-0.0007, 0.0124] |
| Los Angeles (481) | -0.0568 [-0.0674, -0.0465] | 0.0102 [0.0055, 0.0150] | 0.0098 [0.0059, 0.0140] |
| Miami (439) | -0.0071 [-0.0126, -0.0014] | 0.0020 [-0.0023, 0.0065] | 0.0034 [0.0000, 0.0068] |
| Phoenix (126) | -0.0024 [-0.0121, 0.0071] | 0.0086 [0.0007, 0.0172] | 0.0071 [0.0005, 0.0144] |

*Note.* Brier loss differences with pointwise 95% paired seven-event bootstrap intervals. Counts appear in parentheses. Windows share the same full evaluation sample.

**Table C4**
*Rolling 40; original evaluation subset*

| City | Public minus NBM | Public minus Kalshi | Public minus blend |
|---|---|---|---|
| New York (138) | -0.0109 [-0.0185, -0.0027] | 0.0058 [-0.0023, 0.0140] | 0.0059 [-0.0005, 0.0122] |
| Los Angeles (481) | -0.0568 [-0.0675, -0.0466] | 0.0102 [0.0055, 0.0150] | 0.0098 [0.0059, 0.0139] |
| Miami (439) | -0.0071 [-0.0128, -0.0015] | 0.0020 [-0.0021, 0.0064] | 0.0034 [0.0000, 0.0067] |
| Phoenix (126) | -0.0024 [-0.0123, 0.0069] | 0.0086 [0.0007, 0.0169] | 0.0071 [0.0004, 0.0144] |

*Note.* Brier loss differences with pointwise 95% paired seven-event bootstrap intervals. Counts appear in parentheses. Windows share the same full evaluation sample.

**Table C5**
*expanding; full usable*

| City | Public minus NBM | Public minus Kalshi | Public minus blend |
|---|---|---|---|
| New York (138) | 0.0042 [-0.0021, 0.0105] | 0.0208 [0.0133, 0.0286] | 0.0207 [0.0135, 0.0282] |
| Los Angeles (481) | -0.0577 [-0.0675, -0.0486] | 0.0092 [0.0049, 0.0136] | 0.0086 [0.0055, 0.0120] |
| Miami (439) | -0.0097 [-0.0142, -0.0053] | -0.0006 [-0.0042, 0.0029] | 0.0016 [-0.0009, 0.0041] |
| Phoenix (126) | -0.0058 [-0.0124, 0.0007] | 0.0053 [-0.0008, 0.0112] | 0.0050 [-0.0002, 0.0101] |

*Note.* Brier loss differences with pointwise 95% paired seven-event bootstrap intervals. Counts appear in parentheses. Windows share the same full evaluation sample.

**Table C6**
*expanding; original evaluation subset*

| City | Public minus NBM | Public minus Kalshi | Public minus blend |
|---|---|---|---|
| New York (138) | 0.0042 [-0.0021, 0.0106] | 0.0208 [0.0136, 0.0286] | 0.0207 [0.0136, 0.0284] |
| Los Angeles (481) | -0.0577 [-0.0674, -0.0483] | 0.0092 [0.0049, 0.0137] | 0.0086 [0.0054, 0.0120] |
| Miami (439) | -0.0097 [-0.0142, -0.0054] | -0.0006 [-0.0041, 0.0029] | 0.0016 [-0.0008, 0.0040] |
| Phoenix (126) | -0.0058 [-0.0124, 0.0007] | 0.0053 [-0.0006, 0.0111] | 0.0050 [-0.0003, 0.0100] |

*Note.* Brier loss differences with pointwise 95% paired seven-event bootstrap intervals. Counts appear in parentheses. Windows share the same full evaluation sample.